Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions

with the Riemann Hypothesis as a corollary

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"... a 'viewpoint' by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally 'real'. It is when complementary viewpoints of a common reality are conjugated, that is, when our 'eyes' are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ... it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision."

— A. Grothendieck, *Récoltes et Semailles*

"The measure of our success is whether what we do enables people to understand and think more clearly and effectively about mathematics."

— W. Thurston, *On Proof and Progress in Mathematics*

Abstract

Let $\mathcal{R}$ be the commutative ring of slice-preserving slice-regular functions on the compactified octonions $\mathbb{O}^* = S^8$, under the regular $(*)$ product. Part 1 gathers the classical facts about the Riemann zeta function. Part 2 develops the slice-preserving theory, builds $\mathcal{R}$, places the octonionic zeta $\zeta_{\mathbb{O}^*}$ inside it, and proves the zero and equivalence theorems: the classically nontrivial zeros of ζ are residue-$\mathbb{C}$ zero 6-spheres, and the Riemann Hypothesis holds exactly when those spheres are concentric, sharing a single real centre.

Part 3 proves the main result, the *Concentricity Theorem*, for a section $A \in \mathcal{R}$ satisfying four conditions: (C1) meromorphic continuation with a single simple pole, whose pole-cancelled factor continues to a finite nonzero value at p_A; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization whose spherical factors are the residue-$\mathbb{C}$ zeros; and (C4) infinitely many such zeros. The theorem says the residue-$\mathbb{C}$ zero 6-spheres of any such A are concentric.

The proof builds a concentricity detector from the section itself, out of action groupoids, matrix actions, and categorical homotopy theory. Conditions (C1)–(C3) construct one distinguished exponential Möbius action on the projective base $\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$: the continued pole factor's logarithm λ_A is exponentiated in the diagonal matrix $\mathrm{diag}(e^{\lambda_A}, 1)$ and conjugated into the slice Möbius group. The Euler product presents the action on the half-space; condition (C1) continues the pole-cancelled factor through the finite pole p_A, and the

Weierstrass product presents that same continued factor there. The slice-preserving normalization is G_2-equivariant on $\mathbb{O}^*$, and orbit–stabilizer proves that its object and arrow assignments define the sphere-world functor. From then on that functor itself records the directions and concrete Möbius matrix actions, while the A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ has the resulting normalized A-value states and their transports as its fibres. Its Grothendieck construction is $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$.

The residue-$\mathbb{C}$ system is the inverse-image action groupoid $\mathcal{R}_A$ selected inside $\mathcal{A}_A$, with fully faithful inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$, and hence a fully faithful inclusion of totals $\int \iota_A : \int_{\mathcal{B}} \mathcal{R}_A \hookrightarrow \mathcal{T}_A$. For two arbitrary objects of the inverse-image total, the A-specific projective/Möbius transport compares their positioned coordinates and the G_2 fibre arrow compares their slice directions. Hence this inverse-image total is transitive, therefore connected. Its component colimit is consequently a singleton. Applying π_0 fibrewise to $\mathcal{R}_A$, the intrinsic real face of the same bound action descends to a natural transformation $\pi_0 \circ \mathcal{R}_A \Rightarrow \Delta(\mathbb{R})$. Its induced colimit map, evaluated on the constructed zero-sphere representatives, reads one real value $c \in \mathbb{R}$ across the singleton component colimit. Thus every residue-$\mathbb{C}$ zero 6-sphere has centre c. The theorem asserts the shared centre; naming the number is the task of the corollaries.

Downstream corollaries translate residue-$\mathbb{C}$ zeros to classically nontrivial zeros and, for $\zeta_{\mathbb{O}^*}$, use its functional equation to identify $c = \frac{1}{2}$: the Riemann Hypothesis.

Contents

Introduction

The Riemann Hypothesis is usually met head-on, as a question about where the zeros of a single complex function fall along a line. This paper approaches it from the side. It asks instead whether a family of geometric objects, sitting a few dimensions up in the octonions, share a common centre, and shows that they must. The Riemann Hypothesis then follows, not as the thing we set out to prove, but as the shadow this larger fact casts back down onto the complex plane.

A nontrivial zero of the zeta function, seen from the complex line alone, is a lone point in a strip. Seen from the compactified octonions, that same zero opens into a six-dimensional sphere, and the critical-line question becomes a question of arrangement: are these spheres concentric, or scattered into different hyperplanes? It takes the whole continuum of imaginary directions in the octonions, read together, to see whether the alignment holds.

The paper is built to make that convergence visible and, at every step, checkable. Part 1 recalls the classical facts about the Riemann zeta function we will lean on. Part 2 develops the slice-preserving theory of functions on the octonions, builds the commutative ring $\mathcal{R}$ in which the octonionic zeta lives, and proves the reframing at the centre of everything: the classically nontrivial zeros are exactly the residue-$\mathbb{C}$ zero six-spheres, and the Riemann Hypothesis holds precisely when those spheres are concentric. Part 3 proves the main result, the Concentricity Theorem, with a small piece of machinery I think of as a concentricity detector. From the section itself we build one distinguished action on a projective base; orbit–stabilizer proves its slice functor well defined, and the resulting concrete matrix actions let us read off, through categorical homotopy theory, whether the zero-spheres land in a single connected piece. They do; and for the octonionic zeta, that is the Riemann Hypothesis.

No one can picture an eight-dimensional graph in their head, but tools from categorical homotopy theory, like the total-object Grothendieck construction and standard colimit arguments, let us probe this structure and reason about it. The argument is also formalized in Lean, so that each step can be checked; the reader can not only verify the steps but, I hope, come to understand for themselves why the Concentricity Theorem is true and why the Riemann Hypothesis follows.

Part I
Classical Background

1 The Riemann zeta function

The Riemann zeta function is classical, and we take its basic theory as given. We recall only what the construction uses: the meromorphic continuation and functional equation, the Euler product over the primes, the Hadamard product over the zeros, and Hardy's theorem. We then read ζ in compactified form, as a map to the Riemann sphere, which is the form in which it will extend to the octonions.

Theorem 1.1 (Meromorphic continuation and functional equation; Riemann [13]). *The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$: holomorphic as a $\mathbb{C}$-valued function on $\mathbb{C} \setminus \{1\}$, with a single simple pole at $s = 1$. Equivalently, read into the Riemann sphere, the same function is a holomorphic map on the whole complex plane,*

$$\zeta : \mathbb{C} \longrightarrow \mathbb{C}^*, \qquad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \qquad \zeta(1) = \infty,$$

the pole becoming the regular value ∞. Holomorphy is on the domain $\mathbb{C}$ in both readings: with infinitely many zeros, ζ is not rational, so the domain point ∞ carries an essential singularity. (The later extension of the domain *to $\mathbb{C}^*$ assigns a marked value at ∞, Definition 1.6 with Remark 1.7: an assignment of geometry and value, made where the classical statement ends.) The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies*

$$\xi(s) = \xi(1-s), \qquad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ is a nontrivial zero of ζ then so are $\bar{\rho}$, $1-\rho$, and $1-\bar{\rho}$. The trivial zeros of ζ are at $s = -2, -4, \ldots$; the nontrivial zeros lie in the strip $0 < \operatorname{Re} s < 1$. The Riemann Hypothesis asserts that every nontrivial zero has $\operatorname{Re} s = \frac{1}{2}$.

Theorem 1.2 (Infinite Euler product; [15, Ch. 1])**.** *For $\operatorname{Re} s > 1$,*

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - p^{-s}\right)^{-1} = \exp\Big(\sum_{p} \sum_{k \geq 1} \tfrac{1}{k}\, p^{-ks}\Big),$$

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\operatorname{Re} s > 1$.

Theorem 1.3 (Infinite Hadamard product; [15, Ch. 2])**.** *ξ is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros $\{\rho\}$ of ζ, and admits the Hadamard factorization*

$$\xi(s) = \xi(0) \prod_{\rho} \Big(1 - \tfrac{s}{\rho}\Big) e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Corollary 1.4 (Infinitude of the nontrivial zeros)**.** *ζ has infinitely many nontrivial zeros.*

Proof. Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of ζ. □

Theorem 1.5 (Hardy [11])**.** *Infinitely many nontrivial zeros of ζ lie on the line $\operatorname{Re} s = \frac{1}{2}$.*

The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction, ζ is taken in compactified form, with values on the Riemann sphere $\mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2$ (the one-point compactification of $\mathbb{C}$ [12]). The classical facts above are retained as stated.

Definition 1.6 (Compactified classical zeta)**.** $\zeta_{\mathbb{C}^*} : \mathbb{C}^* \to \mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$ for $s \in \mathbb{C} \setminus \{1\}$; $\zeta_{\mathbb{C}^*}(1) = \infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty) := 1$, an assigned marked value (Remark 1.7), selected by the one genuine limit $\zeta(s) \to 1$ as $\operatorname{Re}(s) \to +\infty$ (the p-test: only the $n = 1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C} \setminus \{1\}$, its sole pole being $s = 1$; at the domain point ∞ no analytic claim is made. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s}) = \overline{\zeta_{\mathbb{C}^*}(s)}$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C} \setminus \{1\}$ where $\zeta_{\mathbb{C}^*} = \zeta$.

Remark 1.7 (The value at infinity is a marked point, not an extension)**.** No analytic claim is made, or could be made, at the domain point ∞. A function holomorphic on the whole Riemann sphere is rational, and ζ is not: it has an essential singularity at ∞, tending to 1 as $\operatorname{Re} s \to +\infty$ while passing through the trivial zeros along the negative real axis, so no assigned value makes $\zeta_{\mathbb{C}^*}$ so much as continuous there. Definition 1.6 therefore performs two distinct acts, and only the first is analytic. Regarding the meromorphic ζ as a holomorphic map into the target sphere, with $\zeta_{\mathbb{C}^*}(1) = \infty$, is the classical sphere-valued reading of a meromorphic function. Assigning $\zeta_{\mathbb{C}^*}(\infty) := 1$ supplies the value the compactified setting consumes as data: the point $\infty = N$ is shared by every slice Riemann sphere (Definition 2.4), it is where the pole's value lands, and it is an object of the projective base the transport machinery of Part 3 runs on (Definition 9.23), so compactified value paths are read at it. The assignment is not free. Slice preservation forces the value to lie in every slice sphere at once, and the intersection of the slice spheres is $\mathbb{R} \cup \{N\}$; the limit along the one region through which the construction approaches ∞ — the right half-plane, where $\zeta \to 1$ genuinely — then selects 1. The essential singularity is visible only along directions the construction never travels. What is consumed downstream is the value, never smoothness: no statement in this paper uses continuity or differentiability of $\zeta_{\mathbb{C}^*}$ at ∞. The same discipline governs the whole class of Part 3: an A-section has infinitely many zeros, hence is not rational, hence has an essential singularity at N; accordingly every analytic hypothesis of Definition 10.5 is stated at finite points, and the value at N is carried as bare data (in the Lean development, the field `valueAtInfinity`). This is the marked-extension device of Remark 3.6: the compactification supplies geometry — the pole's value becomes a point of the target sphere, and all slice spheres share the single point N — never an extension. A companion distinction completes the picture. At N, Part 3 positions a group action carrying analytic content: the slice-preserving content the hypotheses supply over the octonions. There N serves as an object of the projective base, as the point of each value sphere at which the distinguished Möbius element arrives in Weierstrass form, and as the shared point of the slice spheres. The analysis that produces the element read there happens at the finite pole — continuation through the simple pole — and its output is positioned at N by the orbit representative. No step of the construction evaluates the section analytically at the domain point ∞: the position and the marked point are one geometric point playing two roles, as a group element's position and as a function's domain point, and only the first is ever active in the construction.

Lemma 1.8 (Compactification preserves the zero set)**.** *$Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.*

Proof. On $\mathbb{C} \setminus \{1\}$ the two functions agree, so they share all zeros there. The two adjoined values lie outside the zero set: $\zeta_{\mathbb{C}^*}(1) = \infty$ is a pole and $\zeta_{\mathbb{C}^*}(\infty) = 1$. Hence compactification preserves the zero set exactly. □

Part II
Slice-Preserving Theory, the Ring $\mathcal{R}$, and the Equivalence

2 Octonions and slices

The octonions are an 8-dimensional real division algebra, the largest of the normed division algebras. This section fixes what we need from them: the algebra itself; Artin's theorem, that any two octonions generate an associative subalgebra, so that powers and slicewise complex analysis make sense; and the complex slices $\mathbb{C}_v$, one for each imaginary direction $v \in S^6$, together with their compactifications, the slice Riemann spheres. All the slices share the real axis and a single point at infinity, the north pole N; that shared geometry is what the later construction turns around.

The octonions $\mathbb{O}$ have basis $\{1, e_1, \ldots, e_7\}$ with each $e_i^2 = -1$.

Definition 2.1. The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

An octonion $s \in \mathbb{O}$ decomposes as $s = \sigma + w$ with $\sigma = \mathrm{Re}(s) \in \mathbb{R}$ and $w = \mathrm{Im}(s) \in \mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$; the unit imaginary octonions form $S^6 \subset \mathrm{Im}(\mathbb{O})$.

Theorem 2.2 (Artin [14])**.** *In an alternative algebra, the subalgebra generated by any two elements is associative.*

Corollary 2.3. *Powers s^n are well-defined for any $s \in \mathbb{O}$.*

Definition 2.4 (Complex slices and slice Riemann spheres)**.** For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$-algebra isomorphism $\varphi_v : \mathbb{C} \to \mathbb{C}_v$, $i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9], holomorphicity on each complex slice separately, as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin's theorem each slice $\mathbb{C}_v$ is an associative subalgebra where classical complex analysis applies unchanged).

Definition 3.1 (Slice regularity; [5, Def. 5.1.1])**.** Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \to \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

Theorem 3.2 (Representation Formula; [5, Thm. 5.1.7])**.** *Let f be slice regular on an axially symmetric slice domain Ω. Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2} v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.*

Theorem 3.3 (Identity Theorem; [5, Cor. 5.1.9])**.** *Let f be slice regular on an axially symmetric slice domain Ω. If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω.*

Theorem 3.4 (Extension Theorem; [5, Thm. 5.1.5])**.** *Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \to \mathbb{C}$ holomorphic. There is a unique slice regular $\mathrm{ext}(g) : \Omega_U \to \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U,\ v \in S^6\}$ with $\mathrm{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0.*

Definition 3.5 (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16])**.** A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

(i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);

(ii) its representation-formula components are $\mathbb{R}$-valued: writing $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ (Theorem 3.2), one has $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$, equivalently the inducing stem function $F = F_1 + \iota F_2$ has $\mathbb{R}$-valued components F_1, F_2;

(iii) f preserves two distinct slices; equivalently $f(\Omega \cap \mathbb{R}) \subseteq \mathbb{R}$ when Ω is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti– Stoppato [7, Rem. 1.8]).

Remark 3.6 (Compactified extension). Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains $\Omega \subseteq \mathbb{O}$ (and $U \subseteq \mathbb{C}$). We use them on the compactified $\mathbb{O}^* = S^8$ and on the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (GPV's slice Riemann manifolds [17]). The passage $\mathbb{O} \rightsquigarrow \mathbb{O}^*$, $\mathbb{C}_v \rightsquigarrow \mathbb{C}_v^*$ adjoins only the shared point $\infty = N$; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

Slice-preserving theory already gives the tools to build the octonionic zeta function. On each slice $\mathbb{C}_v$ we read the compactified classical ζ through the isomorphism $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^*$, and the representation formula glues these slice copies into a single function on all of $\mathbb{O}^*$. The one thing to check is that the two identifications of a slice with $\mathbb{C}$ agree, so that $\zeta_{\mathbb{O}^*}$ is well-defined; they do, because conjugating the input and reversing the identification cancel. The pole of ζ at $s = 1$ is carried to the north pole N. At the shared point N itself the function and the geometry are defined but nothing more is claimed: the value there is the assigned marked value 1 of Remark 1.7 — as real $s \to +\infty$, $\zeta(s) \to 1$, the one genuine limit — and neither continuity nor differentiability at N is asserted or used by the argument.

Definition 4.1 (Octonionic zeta)**.** Here N denotes the *north pole* of S^8: the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty = N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^* = S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \to \mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^* = S_v^2$ (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty) = N$; GPV [17], Prop. 4.1/Thm. 4.2):

(i) for $s \in \mathbb{O} \setminus \mathbb{R}$, with $w = \mathrm{Im}(s)$, $v = w/|w| \in S^6$, $\gamma = |w| > 0$, $\sigma = \mathrm{Re}(s)$, so $s = \sigma + \gamma v$: $\zeta_{\mathbb{O}^*}(s) := \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$;

(ii) for $s \in \mathbb{R} \setminus \{1\}$: $\quad \zeta_{\mathbb{O}^*}(s) := \zeta_{\mathbb{C}^*}(s) = \zeta(s) \in \mathbb{R}$;

(iii) for $s = 1$ (the simple pole, Definition 1.6): $\quad \zeta_{\mathbb{O}^*}(1) := \infty = N$;

(iv) for $s = \infty = N$: $\quad \zeta_{\mathbb{O}^*}(\infty) := \zeta_{\mathbb{C}^*}(\infty) = 1$, the assigned marked value of Remark 1.7: as real $s \to +\infty$, $\zeta(s) \to 1$, and the definition asserts the value only — neither continuity nor differentiability of $\zeta_{\mathbb{O}^*}$ at N is claimed here or used anywhere in the argument.

Proposition 4.2 (Well-definedness)**.** $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \to \mathbb{O}^*$ *is well-defined.*

Proof. For $s \in \mathbb{O} \setminus \mathbb{R}$ the decomposition $s = \sigma + \gamma v$ with $\gamma > 0$, $v \in S^6$ is forced ($\gamma = |\operatorname{Im} s|$, $v = \operatorname{Im} s/|\operatorname{Im} s|$), so the only ambiguity is the identification of the slice $\mathbb{C}_v = \mathbb{C}_{-v}$ with $\mathbb{C}^*$. That slice carries two such identifications, φ_v ($i \mapsto v$) and φ_{-v} ($i \mapsto -v$), under which s reads as $\sigma + i\gamma$ resp. $\sigma - i\gamma$; we must check $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = \varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma - i\gamma))$. By conjugate symmetry $\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$ ([15, Ch. 2]; Definition 1.6), writing $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$ gives $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$: the two sign reversals, one from conjugating the input, one from the reversed identification, cancel. The remaining cases carry no ambiguity: for real $s \neq 1$, $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ for every v; at $s = 1$ the value is the single point $\infty = N$; and at $s = \infty = N$ the value is $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$, again the same for every v since each φ_v fixes $\mathbb{R}$ pointwise. Well-definedness at N is a statement about the value, not about regularity: the assigned marked value 1 lies in $\mathbb{R}$, which every slice sphere contains, so it is unambiguous across all slices, and the function and the geometry are thereby defined at N while continuity and differentiability there are neither claimed nor used (Remark 1.7). □

5 $\mathcal{R}$ is a commutative ring

The slice-preserving sections form a commutative ring $\mathcal{R}$, with $\zeta_{\mathbb{O}^*}$ among its elements. Its multiplication is the regular ($*$) product, which descends from the Cayley–Dickson construction of $\mathbb{O}$ and from slice regularity; on slice-preserving functions it restricts on each slice to the ordinary pointwise product (Wang), and is therefore commutative. The ring axioms, and the integral-domain property, follow slicewise.

We work on $\mathbb{O}^*$; write $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$ for the slice Riemann sphere (Definition 2.4).

Definition 5.1 (The ring of slice-preserving functions under the regular product)**.** $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-preserving functions $f : \mathbb{O}^* \to \mathbb{O}^*$ (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, slice regular away from their marked points — the poles, where the value is ∞, and the shared point N (Remark 1.7) — equipped with pointwise addition and the regular ($*$) product.

Commutativity of $\mathcal{R}$ rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

Theorem 5.2 (Regular product on slice-preserving functions; [16, Rem. 2.11])**.** *For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).*

Proposition 5.3. $(\mathcal{R}, +, *)$ *is a commutative ring with identity* $1 \in \mathcal{R}$.

Proof. Closure under $+$ is pointwise. For $*$: fixing a slice $\mathbb{C}_K$, the product reduces to pointwise multiplication on $\mathbb{C}_K$ by Theorem 5.2, which is commutative because $\mathbb{C}_K$ is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity. □

Proposition 5.4 ($\mathcal{R}$ is an integral domain)**.** *If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.*

Proof. $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$ in $\mathbb{C}_K \cong \mathbb{C}$ (Theorem 5.2). The restrictions $f|_{\Omega_K}$, $g|_{\Omega_K}$ are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes. □

6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and I-independent lift, and the winding lift. Each is established in the literature on the uncompactified $\mathbb{O}$ and used here on $\mathbb{O}^* = S^8$ and the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

Theorem 6.1 (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23])**.** *The exponential* $\exp : \mathbb{O} \to \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k\geq 0} q^k/k!$ *(powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \qquad x, y \in \mathbb{R},\ I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ *depends only on the real part, and* $\exp$ *is nowhere zero.*

Theorem 6.2 (The logarithm manifold $E_{\mathbb{O}}^+$; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3])**.** *Let* $T : \mathbb{O} \to \mathbb{O} \times \operatorname{Im} \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y,\ Iy)$, *and* $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$. *The* logarithm manifold *is* $E_{\mathbb{O}}^+ := T(\mathbb{O}^+)$, *the semi-helicoidal hypercomplex* 8*-manifold; equivalently it is the image of the exponential graph, parametrised by the* $E_{\mathbb{O}}^+$-exponential

$$E : \mathbb{O} \longrightarrow E_{\mathbb{O}}^+ \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \qquad E(x + Iy) = (\exp(x + Iy),\, Iy) = (e^x \cos y + Ie^x \sin y,\ Iy),$$

an immersion and a diffeomorphism, with inverse the $E_{\mathbb{O}}^+$-logarithm $L : E_{\mathbb{O}}^+ \to \mathbb{O}$, $L(q, p) = \log |q| + p$ *(here* $p = \operatorname{Arg}(q)$*). Writing* π *for the projection to the first factor,* $\pi \circ E = \exp$ *(Theorem 6.1). The manifold* $E_{\mathbb{O}}^+$ *is an adapted blow-up of* $\mathbb{O}$ *along the real axis. Its projection* $\pi : E_{\mathbb{O}}^+ \to \mathbb{O}\setminus\{0\}$ *has the degenerate real fibres responsible for the failure of covering and openness. The blow-up unwraps the multivalued logarithm into the single-valued* L*, and a branch of the hypercomplex logarithm is* $\log_{\mathbb{O}} = L \circ (\pi|_\Omega)^{-1}$ *on any path-connected* $\Omega \subset E_{\mathbb{O}}^+$ *on which* $\pi|_\Omega$ *is injective.*

Lemma 6.3 (The degenerate set of the exponential; its fibre over a real value)**.** *With* π *the first-factor projection,* $\pi \circ E = \exp$ ([17, Rem. 5.2(a)], the commuting triangle)*. Its spherical* degenerate set *makes the projection an adapted blow-up with failures of covering and openness* ([17, Rem. 5.2(b)])*. The direction* $I(q)$ *becomes singular at every point of* $\mathbb{R}$ ([21, Rem. 2.1])*.*

Proved from the slice form. *For* $r > 0$*,*

$$\exp^{-1}(-r) = \big\{ \log r + I(2k+1)\pi :\ I \in S^6,\ k \in \mathbb{Z} \big\} :$$

by Theorem 6.1, $\exp(x+Iy) = e^x \cos y + Ie^x \sin y = -r$ *forces* $e^x \sin y = 0$*, hence* $y \in \pi\mathbb{Z}$*; negativity of the value forces* $y = (2k+1)\pi$ *and* $x = \log r$*; the direction* I *is unconstrained. The fibre is thus indexed by the single real* level $\log r = \log |-r|$ *and, within it, by the odd winding indices* $2k+1$ *carried as band data (the winding lift, Theorem 6.5; [21, Cor. 5.21]). The fibre formula also appears as Preface motivation in [17] (p. 972); the slice-form derivation above proves it.*

Definition 6.4 (Slice restriction and the I-independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2])**.** A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \qquad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ of Definition 2.4. The *I-independent lift* is the fact that this holomorphic stem is the *same* for every v: there is one holomorphic $A_\circ : \mathbb{C}^* \to \mathbb{C}^*$ with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_\circ \circ \varphi_v^{-1} \qquad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \to \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$-valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad F_1, F_2 \ \mathbb{R}\text{-valued}.$$

It is this single I-independent slice stem that makes a slice-preserving section G_2-equivariant, hence blind to the G_2-orbit direction (Definition 9.2).

Theorem 6.5 (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11])**.** *Let $\gamma : [a, b] \to \mathbb{O}\backslash\{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation).* A continuation of the hypercomplex logarithm along γ *is a path $\tilde{\gamma} : [a, b] \to \mathbb{O}$ with $\exp \circ \tilde{\gamma} = \gamma$, and a* lift of γ *to the logarithm manifold $E_{\mathbb{O}}^+$ (Theorem 6.2) is a path $\Gamma : [a, b] \to E_{\mathbb{O}}^+$ with $\mathrm{pr}_1 \circ \Gamma = \gamma$:*

$$\begin{tikzcd} & \mathbb{O} \\ {[a,b]} \arrow[ur, "\tilde{\gamma}"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \arrow[u, "\exp"'] \end{tikzcd} \qquad \begin{tikzcd} & E_{\mathbb{O}}^+ \arrow[d, "\mathrm{pr}_1"] \\ {[a,b]} \arrow[ur, "\Gamma"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

The two data are equivalent*: a lift Γ exists if and only if a continuation $\tilde{\gamma}$ exists, and they correspond by*

$$\tilde{\gamma} = L \circ \Gamma, \qquad \Gamma = (\exp \circ \tilde{\gamma},\ \mathrm{Im}\, \tilde{\gamma}),$$

where L is the $E_{\mathbb{O}}^+$-logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop *$\gamma : S^1 \to \mathbb{O} \setminus \{0\}$ the lift is taken on the universal cover* $\exp : i\mathbb{R} \to S^1$*, as a continuous $\Gamma : i\mathbb{R} \to E_{\mathbb{O}}^+$ with $\mathrm{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):*

$$\begin{tikzcd} i\mathbb{R} \arrow[r, "\Gamma"] \arrow[d, "\exp"'] & E_{\mathbb{O}}^+ \arrow[d, "\mathrm{pr}_1"] \\ S^1 \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

Proposition 6.6 (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22])**.** *The lift of Theorem 6.5 is governed by the companion structure of γ.*

(i) Uniqueness (tameness). *γ is* tame *when it admits a* unique *companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).*

(ii) Existence and closure. *A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and* if it exists, the lift is itself a loop *([21, Cor. 5.13]; pinned against the arXiv text; the earlier citation to Cor. 5.22 is superseded).*

(iii) Winding number. *For an untwisted tame loop with even circular signature σ^c, the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).*

7 Slice preservation and symmetries

With the ring in hand, we check that $\zeta_{\mathbb{O}^*}$ actually lives in it, and record the symmetry that will organize its zeros. Slice preservation says $\zeta_{\mathbb{O}^*}$ keeps each slice inside itself; together with slice regularity this makes it a section of $\mathcal{R}$. The symmetry is the action of the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$, which fixes the real axis and rotates the imaginary directions S^6, and $\zeta_{\mathbb{O}^*}$ is equivariant under it. This forces every non-real zero to fill a whole 6-sphere, the geometry the next section reads off.

Theorem 7.1 (Slice preservation). *$\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.*

Proof. Fix $v \in S^6$ and $s \in \mathbb{C}_v^*$. If $s \in \mathbb{C}_v \setminus \mathbb{R}$ then $s = \sigma \pm \gamma v$ and $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$ (Definition 4.1(i)). If $s \in \mathbb{R}\setminus\{1\}$ then $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$; while $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$ and $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$. In every case the value remains in $\mathbb{C}_v^*$. □

Theorem 7.2 ($\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$). $\zeta_{\mathbb{O}^*} \in \mathcal{R}$.

Proof. On each slice the restriction is the compactified classical zeta read through the isomorphism φ_v, namely $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$, holomorphic into $\mathbb{C}_v^*$ away from the single pole. Equivalently $\zeta_{\mathbb{O}^*} = \mathrm{ext}(\zeta)$ is the slice-regular extension of the holomorphic, conjugation-symmetric ζ on $\mathbb{C} \setminus \{1\}$ (Theorem 3.4, [5, Thm. 5.1.5]); hence $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (slice regularity is slicewise holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving, $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$. A slice-regular, slice-preserving function $\mathbb{O}^* \to \mathbb{O}^*$ is by definition a section of $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$ (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Its zeros (Theorem 8.2) make $\zeta_{\mathbb{O}^*}$ a nonunit. □

Definition 7.3 (The automorphism group G_2). $G_2 = \mathrm{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$-algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\mathrm{Im}\,\mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g\cdot(\sigma+\gamma v) = \sigma + \gamma\, g(v)$.

Theorem 7.4 (G_2-equivariance). *$\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.*

Proof. For $s \in \mathbb{R}$, $g \cdot s = s$ and $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$ is fixed, so both sides equal $\zeta_{\mathbb{O}^*}(s)$. For a non-real $s = \sigma + \gamma v$ ($v \in S^6$, $\gamma > 0$), g carries the slice $\mathbb{C}_v$ to the slice $\mathbb{C}_{g(v)}$ by the isometric relabelling $\varphi_{g(v)} = g \circ \varphi_v$ (Definition 7.3). Since $\zeta_{\mathbb{O}^*}$ is defined slicewise by the same $\zeta_{\mathbb{C}^*}$ read through φ_v (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}\big(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)\big) = g\big(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))\big) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

□

Theorem 7.5 ([3]). *G_2 acts transitively on S^6 with stabilizer $\mathrm{SU}(3)$: $\mathrm{SU}(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.*

Remark 7.6 (The G_2-action extends to $\mathbb{O}^* = S^8$). The action above is stated on $\mathbb{O}$, but $\mathcal{H}_1$ (Part 3) runs on the compactified $\mathbb{O}^* = S^8$. Each $g \in G_2$ fixes the real axis pointwise (an automorphism fixes 1, hence all of $\mathbb{R}$) and acts on $\mathrm{Im}\,\mathbb{O} = \mathbb{R}^7$ as an element of $SO(7)$; in particular g is a linear isometry of $\mathbb{R}^8 = \mathbb{O}$, hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ fixing the point at infinity $\infty = N$ (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus $G_2 \curvearrowright \mathbb{O}^*$ fixes the compactified real axis $\mathbb{R} \cup \{\infty\}$, a great circle through N, pointwise, and acts on each direction-sphere S^6 exactly as in Theorem 7.5. This is the action on which $\mathcal{H}_1$ is built, with $\mathcal{R}$ its ring of invariants over $\mathbb{O}^*$.

8 Zero geometry and the equivalence

This section is the bridge the whole paper is built to cross. It shows that the zeros of $\zeta_{\mathbb{O}^*}$ are exactly the classical nontrivial zeros, now spread out: each nontrivial zero ρ of ζ becomes a whole 6-sphere of zeros of $\zeta_{\mathbb{O}^*}$, its residue-$\mathbb{C}$ zero-sphere, sitting at real part $\operatorname{Re}\rho$. The Riemann Hypothesis, that every ρ has real part $\frac{1}{2}$, becomes the statement that all these 6-spheres share the same real part, that they are concentric. From here on we work with the spheres.

Theorem 8.1 (Zero Equivalence Theorem). *Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,*

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \iff \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.8 these are exactly the nontrivial zeros of the classical ζ.

Proof. By Definition 4.1(i), $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$. The slice identification $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^* = S_v^2$ is an isomorphism of slice Riemann spheres (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended by $\varphi_v(\infty) = N$; GPV [17]). An isomorphism is injective and sends 0 to 0, so $\varphi_v(z) = 0 \iff z = 0$; with $z = \zeta_{\mathbb{C}^*}(\rho)$ this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in $\mathbb{C}_v^*$ so that φ_v^{-1} applies on the non-real side.) The final clause is Lemma 1.8. □

Theorem 8.2 (Nontrivial zeros are infinitely many disjoint 6-spheres). *For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set*

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma\, S^6 \subset \mathbb{O}.$$

Then:

- *(i)* S_ρ is *a* 6*-sphere, the round sphere of radius γ centred at the real point σ, equivalently the G_2-orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser* SU(3)*, Theorem 7.5, Baez [3]).*
- *(ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar\rho}$;*
- *(iii) the nontrivial-zero set is the disjoint union $Z^{\mathrm{nt}}_{\mathbb{O}^*} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar\rho\}$, with distinct pairs giving disjoint spheres;*
- *(iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).*

Proof. (i) The map $v \mapsto \sigma + \gamma v$ is the affine map $x \mapsto \sigma + \gamma x$ of $\operatorname{Im}\mathbb{O} \cong \mathbb{R}^7$ restricted to S^6; with $\gamma > 0$ its image is the round 6-sphere of radius γ centred at σ. By G_2-equivariance (Theorem 7.4) and transitivity of G_2 on S^6 (Theorem 7.5), $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma\, g(v_0) : g \in G_2\} = S_\rho$. So S_ρ *is* S^6, scaled by γ, translated to σ, realized as a single G_2-orbit; no embedding or diffeomorphism is invoked.

(ii) For any $v \in S^6$, $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$ by the Zero Equivalence Theorem 8.1. And $S_{\bar\rho} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$, since $v \mapsto -v$ is a bijection of S^6.

(iii) For "$\subseteq$": a non-real zero $s = \sigma + \gamma v$ of $\zeta_{\mathbb{O}^*}$ forces $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$ by Theorem 8.1, so $s \in S_\rho$ with $\rho = \sigma + i\gamma$ nontrivial (Lemma 1.8); "$\supseteq$" is (ii). Disjointness: if $\sigma + \gamma v = \sigma' + \gamma' w$, uniqueness of the real/imaginary parts gives $\sigma = \sigma'$ and $\gamma v = \gamma' w$, whence $\gamma = \gamma'$ (taking norms, $|v| = |w| = 1$) and $\{\rho', \bar\rho'\} = \{\rho, \bar\rho\}$; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). □

Theorem 8.3 (The equivalence: concentricity $\Leftrightarrow$ RH)**.** *The following are equivalent:*

(a) the Riemann Hypothesis;

(b) the nontrivial-zero 6*-spheres* $\{S_\rho\}$ *of* $\zeta_{\mathbb{O}^*}$ *(Theorem 8.2) are* concentric*: they share a single common centre, necessarily a point of the real axis.*

When they hold, the common centre is $\frac{1}{2}$*.*

Proof. Each S_ρ is centred at the real point σ_ρ, where $\rho = \sigma_\rho + i\gamma_\rho$ (Theorem 8.2(i)).

$(a) \Rightarrow (b)$: under RH every $\sigma_\rho = \frac{1}{2}$, so all spheres share the centre $\frac{1}{2}$.

$(b) \Rightarrow (a)$: suppose the spheres share a common centre $c \in \mathbb{R}$. By the functional equation (Theorem 1.1), if ρ is a nontrivial zero then so is $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$, whose sphere $S_{1-\bar{\rho}}$ is centred at $1 - \sigma_\rho$. Concentricity forces $\sigma_\rho = c = 1 - \sigma_\rho$, hence $\sigma_\rho = \frac{1}{2}$; as ρ was arbitrary this is RH, and the common centre is $\frac{1}{2}$. ☐

Part III
The Categorical Construction and the Concentricity Theorem

Part 2 revealed that ζ carries a special geometry: spread onto the octonions, its zeros are concentric exactly when the Riemann Hypothesis holds. That raises the question this part answers. What general properties of a section of the ring $\mathcal{R}$ of slice-preserving functions force its residue-$\mathbb{C}$ zeros to be concentric? Following that question leads to four conditions, C1–C4, and to the whole class of functions satisfying them, which we call the A-sections. The main result, the Concentricity Theorem, is that the residue-$\mathbb{C}$ zeros of any A-section are concentric. The octonionic zeta is one A-section among many, and the Riemann Hypothesis is one of its corollaries.

From the section's own data we read one distinguished exponential Möbius action on the Riemann sphere, presented by the Euler product on the half-space and by the Weierstrass product after continuation through the finite pole p_A; orbit–stabilizer proves once that the resulting object and arrow assignments define a functor to SphereWorld. Thereafter its concrete projective/Möbius matrix actions, together with the same action on normalized A-value states, assemble the functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. Its Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ gathers every state and every value-transport into one total object. The zero-sphere system is the inverse-image subdiagram $\mathcal{R}_A$, included by $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$; its action morphisms, the concrete projective/Möbius matrices in the base and the G_2 matrices in the fibres, make $\int_{\mathcal{B}} \mathcal{R}_A$ transitive and hence connected. The component colimit therefore collapses to a singleton, and its real-value readout is one centre $c \in \mathbb{R}$. The concentricity proof takes place in this residue-$\mathbb{C}$ inverse-image system inside the ambient total.

9 The geometric worlds and the categorical construction

This section assembles the parts the detector runs on. Part 2 established that exponentials are slice-preserving and live in the ring $\mathcal{R}$; this section develops a theory for working with these functions, one that shows how they *act*, from the point of view of action-groupoid theory. The key categories, the ones that capture the analysis and the geometry of these actions, are three: the octonionic world $\mathcal{H}_1$; the projective base $\mathcal{B}$, by the slice-preserving literature the home of the unique tame continuous lift of the degenerate GPV fibre; and a world of Riemann spheres, a whole continuum of $\mathbb{C}_I^*$, each with its own Möbius self-maps. The general exponential slice-preserving action, and the A-section-specific action the proof uses, naturally carry value states of $\mathbb{O}^*$ to value states of $\mathbb{O}^*$; the theory requires slice-preserving normalization, so a functor on $\mathbb{O}^*$, the A-section equivariant functor below, assembles what such a total value system looks like: the total Grothendieck construction. To get there, we first explain the theory minimally, for one exponential action: orbit–stabilizer makes the main slice-preserving functor well defined, and the one construction is simultaneously a value state, a group element, an action, and a functor. The resulting functor then supplies the concrete matrix actions used in the total Grothendieck construction and in the transitivity proof for the A-section's residue-$\mathbb{C}$ subsystem.

The categorical inventory is small, and we lay it out once. Five standing groupoids are introduced: the octonionic world $\mathcal{H}_1$, the sphere world, the projective base $\mathcal{B}$, the total $\mathcal{T}_A$, and the residue-$\mathbb{C}$ total $\int_{\mathcal{B}} \mathcal{R}_A$, with the G_2 fibre groupoids $\mathcal{A}_A(b)$ and $\mathcal{R}_A(b)$ standing over the base inside the two totals. Three functors run between them: the equivariant realization $A_{\mathbb{O}}$, the slice functor A^{slice}, and the A-section functor $\mathcal{A}_A$. The slice functor depends on the disk action, and orbit–stabilizer proves its assignments well defined, so the paper first explains how that slice-preserving functor is built. There is a general theory here, slice-preserving ring action groupoids over $\mathbb{O}^*$, one for every element of $\mathcal{R}$; this paper builds the A-section instance, because the hypotheses select one element. The point of building functors is the tools they unlock: the total Grothendieck construction is what this proof uses, and other tools of action-groupoid theory and categorical homotopy stand ready. Thomason's theorem [25] places the Grothendieck construction in its homotopy-colimit context. The proof developed here runs through ordinary connected components: the hypotheses assemble the A-section-specific slice-preserving total, its inverse residue-$\mathbb{C}$ system is transitive, transitivity makes it connected, connectedness makes π_0 a singleton. Applying the Grothendieck-components theorem to the constructed functor $\mathcal{R}_A$ identifies that singleton with $\operatorname{colim}_{\mathcal{B}}(\pi_0 \circ \mathcal{R}_A)$. The intrinsic real face of the bound A-specific action descends through the fibrewise components to a natural transformation $\pi_0 \circ \mathcal{R}_A \Rightarrow \Delta(\mathbb{R})$, whose induced map from that singleton colimit reads the one real number.

What would the graph of a slice-preserving section look like? For an ordinary function one draws the set of pairs (input, value). Here input and value both live on the compactified octonions, so the graph sits in sixteen real dimensions, and no one will ever see it. But slice preservation organizes it completely: the value at a point lies on that point's own slice sphere, and one holomorphic stem is repeated in every direction. The graph is one object in the octonions, and each direction $I \in S^6$ shows it as an ordinary graph: the slice Riemann sphere $\mathbb{C}_I^*$ with the pairing (input coordinate, value coordinate) of the restricted stem, the same picture in every direction, glued along the shared real circle and the shared point N. The construction of this section is that description made categorical, and it is the situation the Grothendieck construction was made for: since the fibered categories of SGA1 it has held one total object as compatibly glued fibres over a base. A state will record an input coordinate together with its positioned image and the value realized there, a point of the graph made into an object, and the groupoids will record the symmetries that carry graph points to graph points: projective transformations of the shared real circle, Möbius motions within each sphere, and G_2 rotations of the direction. A reader who keeps "one graph in the octonions, seen one slice sphere at a time" in mind has the right picture for everything below.

The compactified octonions and the point at infinity

Definition 9.1 (The compactified octonions $\mathbb{O}^* = S^8$)**.** Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$; the slice Riemann-sphere structure on S^m, $m =$

$\dim_{\mathbb{R}} K \in \{4, 8\}$, is GPV [17], Prop. 4.1 and Thm. 4.2. The adjoined point is the north pole N; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`, $\mathbb{O}^* =$ `OnePoint` $\mathbb{O}$ with $\infty =$ `OnePoint.infty`, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$, the point at infinity $\infty = N$. The groupoids $\mathcal{H}_1$ and SphereWorld are built on these compactified values (Definition 9.3); the distinct projective base $\mathcal{B}$ and direction-indexed sphere world used in the construction below are introduced in Definition 9.23.

There is one geometric north pole $N = \infty$, and it plays a role in each groupoid. In $\mathcal{H}_1$ this point of $\mathbb{O}^*$ is a G_2-fixed object: its own connected component, with $\mathrm{Aut}(N) = G_2$. In SphereWorld it is the common point at infinity of the slice Riemann spheres, fixed by every direction arrow. The section relates the two roles through C1: the pole of A has value ∞, so the section carries the pole point to N on every slice sphere at once. The residue field of a closed zero sorts the picture: residue-$\mathbb{R}$ (the G_2-fixed locus $\mathbb{R} \cup \{N\}$, classically trivial) versus residue-$\mathbb{C}$ (the 6-sphere orbits, classically nontrivial; the classification is Part 2 and Theorem 8.2). The later total construction uses this one shared compactified point N, the same for every zero index.

Looking forward: everything the construction does at N happens in action groupoids: N is an object, and the arrows are given by group elements, the distinguished Möbius action and G_2. N serves as a base object, as the shared point of every slice sphere, and as the value of the section at its pole; the section's own value at the domain point N enters as the marked datum of Remark 1.7. The infinite Euler family of C2 with the infinite divisor of C3–C4 makes every A-section essentially singular at the domain point N; what Part 3 reads at N is what the construction supplies there — the geometry of the point and the marked value.

What a slice-preserving section does

Definition 9.2 (The section's slice data and equivariance)**.** Let $A \in \mathcal{R}$. By the I-independent lift (Definition 6.4; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$-valued components,

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad F_1, F_2 \ \mathbb{R}\text{-valued},$$

the same for every $I \in S^6$. Three consequences used below:

(i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I: the value at a point lies on that point's own slice sphere (Definition 5.1).

(ii) *G_2-equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I)\, F_2 = g \cdot \big(F_1 + I\, F_2\big) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2-fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i).

(iii) *Blindness to the sphere direction.* On a G_2-orbit $S_{(\sigma,\gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

A slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2-morphisms already present in both worlds.

The octonionic slice-preserving action groupoids

Three groups act on this picture, and they are the only three. $PGL(2, \mathbb{R})$ acts on the compactified real axis all slices share — projective transformations of the one great circle. $G_2 = \mathrm{Aut}(\mathbb{O})$ acts on the directions: it fixes the real axis pointwise and rotates S^6 transitively. And each slice Riemann sphere carries the Möbius transformations of its own compactified plane. A group action is naturally a groupoid, and this is the only categorical device the paper needs: the *translation groupoid* of $G \curvearrowright X$ has the points of X as objects and the elements $g \in G$ satisfying $g \cdot x = y$ as its morphisms $x \to y$, so an arrow *is* one group element g with $g \cdot x = y$. The name earns its keep for one reason: $\pi_0(G \ltimes X) = X/G$ recovers the orbits (Quillen [22, §1]; in Lean, `CategoryTheory.ActionCategory`), and the readout of the main theorem uses exactly this. Every carrier below is a groupoid of one of these three actions. The sphere world SphereWorld is the direction-indexed continuum of those action groupoids: one Riemann sphere per direction, each carrying its own Möbius action, with the G_2 arrows moving between sphere directions. This structure is defined explicitly in the next subsection: SphereWorld holds a continuum of Riemann-sphere pictures side by side, one for every direction — the multiplied viewpoints of the epigraph, and the form in which an eight-dimensional graph lets itself be seen. Concretely, the projective action will be read in this direction-indexed sphere-world groupoid: its objects are directions $I \in S^6$ equipped with their compactified slice spheres, its morphisms carry one G_2 automorphism and one Möbius transformation, and a normalized value state pairs such a direction with a point of its slice.

Definition 9.3 (The octonionic world and the slice decomposition)**.** The octonionic world is

$$\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*.$$

Its components are the G_2-orbits: the compactified real fixed locus $\mathbb{R} \cup \{N\}$ and, for each $\sigma \in \mathbb{R}$ and $\gamma > 0$, the residue-$\mathbb{C}$ sphere $S_{(\sigma,\gamma)} = \{\sigma + \gamma v : v \in S^6\}$, labelled by its real centre σ and radius γ (Baez [3]; Theorem 7.5).

SphereWorld organizes the same compactified values by their slice decomposition. Every slice contains the real axis, and

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \qquad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

Its objects are the directions $I \in S^6$ with their compactified slice spheres — one sphere per direction, each with the Möbius action of its own compactified plane, the G_2 arrows relabelling which sphere one stands on; the points 0 and N lie on every slice at once. Möbius transformations thus enter the construction twice: complex ones as each slice sphere's own action in SphereWorld, and real ones as the projective base below, $\mathrm{PGL}(2, \mathbb{R})$ acting on the one compactified real line all slices share.

Slice preservation itself determines what must be tracked: the projective great circle every slice shares — the base $\mathcal{B}$ below; the continuum of Riemann spheres — SphereWorld; and the octonionic world $\mathcal{H}_1$, where the image of a slice-preserving function assembles them.

The projective base

The base is the projective action groupoid

$$\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R}).$$

Its objects are points of the compactified real projective line, the real great circle of $\mathbb{O}^*$: the slice through the real axis that every slice sphere shares. Its arrows are projective elements: an arrow $h : x \to y$ is the class of an invertible real matrix acting on the compactified line,

$$h = \left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right], \qquad h \cdot x = \frac{ax+b}{cx+d} = y,$$

one element of $\mathrm{PGL}(2,\mathbb{R})$ carrying x to y. We call these projective elements throughout.

The sphere world

The sphere world is the companion category to the base (`SphereWorld`), and it is the one category of this paper with a *continuum* of groups. Its objects are a continuum of Riemann spheres inside $\mathbb{O}^*$: one compactified slice sphere $\mathbb{C}_I^* \subset \mathbb{O}^*$ for each unit imaginary direction $I \in S^6 \subset \mathbb{O}$, each carrying the Möbius group of its own compactified plane: the invertible complex matrices modulo the scalar matrices,

$$\mathrm{M\ddot{o}b}(\mathbb{C}^*) = \mathrm{GL}(2,\mathbb{C})/\{\lambda\mathbf{1} : \lambda \in \mathbb{C}^\times\} = \mathrm{PGL}(2,\mathbb{C}),$$

with $\mathbf{1}$ the identity matrix. An arrow is a pair: one automorphism in G_2, relabelling which sphere one stands on, and one Möbius transformation, acting on the sphere; a normalized value state pairs a direction with a point of its slice. Beside the base the picture is complete: the base is the real Möbius world of the one line all slices share, and the sphere world is the continuum of complex Möbius worlds, one for each unit imaginary direction.

The disk action: a general slice-preserving action from the base to the sphere world

The aim is to understand slice-preserving maps on $\mathbb{O}^*$ in terms of action groupoids, and the two groupoids are already on the table: the projective base $\mathcal{B}$ just defined, whose arrows are classes of real matrices, and the sphere world just introduced, whose spheres each carry their own complex Möbius group. This subsection builds the generic material of an assignment from the first to the second: one complex Möbius element, produced by exponentiation and read onto every slice sphere by conjugation — a slice-preserving action. Everything here is general, with no A-section hypothesis used; the hypotheses are stated in the next subsection, and they specialize this general functor to the A-section's slice functor A^{slice} below. The object being built is a functor from $\mathcal{B}$ to the sphere world — an assignment of objects to objects and arrows to arrows, respecting sources and targets, that carries identities to identities and composites to composites; these two equalities are its *functoriality* — and the values of a section already live in the sphere world: each value sits on its point's own slice sphere. Two groups are already chosen, and they are the only two. $\mathrm{PGL}(2,\mathbb{R})$ is the classes of invertible real matrices: the arrows of the base, acting on the compactified real line that every slice shares. $\mathrm{GL}(2,\mathbb{C})$ is the invertible complex matrices: its classes are the Möbius transformations, one group on each slice sphere, acting on that sphere's compactified plane. The real classes are already the real part of each Möbius group, and the disk automorphisms, the Blaschke maps, are their Cayley conjugates. This is what lets the main general slice-preserving exponential functor be built from a general *analytic* ring element equipped with its degenerate GPV base. The construction uses two matrices, the Cayley matrix and the exponential:

$$C = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}, \qquad M(z) = \begin{pmatrix} e^z & 0 \\ 0 & 1 \end{pmatrix}:$$

C is the map $w \mapsto (w-i)/(w+i)$, carrying the base's compactified real line onto the unit circle of a slice sphere; M takes a complex number z to the diagonal matrix whose entry is e^z, exponentiation in matrix form. An invertible matrix acts on the Riemann sphere by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2,\mathbb{C}), \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot w = \frac{aw+b}{cw+d}.$$

The class of the matrix up to scalars acts, and this is the Möbius map itself: a scalar multiple, all four entries scaled at once, enters numerator and denominator together and cancels,

$$\begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix} \cdot w = \frac{\lambda a\, w + \lambda b}{\lambda c\, w + \lambda d} = \frac{\lambda\,(aw+b)}{\lambda\,(cw+d)} = \frac{aw+b}{cw+d}, \qquad \lambda \in \mathbb{C}^{\times}.$$

The action is total on the compactified plane: numerator and denominator never vanish at the same w, for if both did, then $a(cw+d) - c(aw+b) = ad - bc$, the determinant, would vanish; where the denominator alone vanishes the value is the point at infinity N. And it is a group action: substituting one fraction into the other multiplies the matrices,

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \left(\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \cdot w \right) &= \frac{a\,\frac{a'w+b'}{c'w+d'} + b}{c\,\frac{a'w+b'}{c'w+d'} + d} = \frac{a(a'w+b') + b(c'w+d')}{c(a'w+b') + d(c'w+d')} \\ &= \frac{(aa'+bc')\,w + (ab'+bd')}{(ca'+dc')\,w + (cb'+dd')} = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \right) \cdot w, \end{aligned}$$

the middle step clearing $c'w + d'$ from numerator and denominator. Acting twice is acting by the matrix product, and every later appeal to the group action is an appeal to this computation.

The functor being built must act on any slice sphere, at any normalization, so the first fact to establish is that the Möbius action reaches every point of a sphere.

Lemma 9.4 (The Möbius action is transitive)**.** *Any two points of a slice Riemann sphere are joined by one Möbius transformation: the action of the classes of* $\mathrm{GL}(2,\mathbb{C})$ *on the compactified plane is transitive.*

Proof. The translation to t and the inversion are the classes

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \cdot w = w + t, \qquad s = \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right], \quad s \cdot w = \frac{1}{w}.$$

The translation by t carries 0 to t: one matrix for each finite value. The inversion at 0 has numerator $0 \cdot 0 + 1 = 1$ and denominator $1 \cdot 0 + 0 = 0$: the denominator alone vanishes, and where that happens the value is the point at infinity (the totality rule above). So $s \cdot 0 = N$. The action at N is then computed by the displayed group-action calculation:

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot N = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot (s \cdot 0) &= \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \cdot 0 \\ &= \begin{pmatrix} b & a \\ d & c \end{pmatrix} \cdot 0 = \frac{a}{c}, \end{aligned}$$

the first equality by $N = s \cdot 0$, the second by the displayed group-action calculation, the third by the matrix product — the point N itself when $c = 0$. So the action is transitive: any two points of the sphere are joined by one class, a composite of the displayed translations and inversion. □

The real projective action supplies the more specific comparison used later. Let $a, b \in \mathbb{R}$ and let u, v lie in the same open half-plane. The real Möbius transformation

$$z \longmapsto b + \left(\frac{\mathrm{Im}(1/(v-b))}{\mathrm{Im}(1/(u-a))} \, \frac{1}{z-a} + \mathrm{Re}(1/(v-b)) - \frac{\mathrm{Im}(1/(v-b))}{\mathrm{Im}(1/(u-a))} \, \mathrm{Re}(1/(u-a)) \right)^{-1} \tag{PGL}$$

is represented by an invertible real matrix. Its positive real ratio is well defined because the two imaginary parts have the same sign. At $z = a$ the inner expression is the point at infinity, so the displayed transformation has value b. At $z = u$ the inner expression is exactly $1/(v-b)$, so its value is v. Thus one concrete element of $\mathrm{PGL}(2, \mathbb{R})$ simultaneously carries the base point a to b and the half-plane point u to v. Conjugation by cayleyProjective gives the corresponding matrix action on every slice Riemann sphere. This calculation will be cited when the residue total is proved transitive.

The diagonal matrix acts by

$$M(z) \cdot w = e^z w = e^x \big(e^{iy} w\big), \qquad z = x + iy,$$

the modulus scaled by e^x and the argument advanced by y.

Now the orbit–stabilizer construction. Take a value w somewhere on a slice Riemann sphere; by the transitivity just shown it lies in the orbit of the Möbius action, and in particular C is itself an invertible class, so there is exactly one point w_0 with

$$w = C \cdot w_0, \qquad w_0 = C^{-1} \cdot w,$$

the value standing as the C-positioned image of w_0. The matrix $M(z)$ acts at w_0 by $M(z) \cdot w_0 = e^z w_0$; to act on the value w where it stands, conjugate, and group multiplication computes the conjugated element doing exactly that:

$$(C\, M(z)\, C^{-1}) \cdot w = (C\, M(z)\, C^{-1}) \cdot (C \cdot w_0) = C \cdot \big(M(z) \cdot w_0\big),$$

the inner $C^{-1}C$ cancelling by associativity of the group action. This is orbit–stabilizer, and it is the positioning pattern of the whole paper: to act at a positioned location, conjugate the action by the element that does the positioning — it runs over the base below at every object and arrow. The operation just performed gets the name it keeps.

Definition 9.5 (The Cayley conjugation and the disk action)**.** The passage from a complex number to a slice Möbius map is two steps, exponentiate then conjugate:

$$z \;\longmapsto\; M(z) \;\longmapsto\; D(z) = \mathrm{cayleyProjective}\big(M(z)\big) = \big[C\, M(z)\, C^{-1}\big] \in \mathrm{M\ddot{o}b}(\mathbb{C}^*),$$

and cayleyProjective $= \big[\, C\,(\cdot)\, C^{-1} \,\big]$ is this same conjugation on every invertible matrix of $\mathrm{GL}(2, \mathbb{C})$.

cayleyProjective takes any invertible matrix, because the positioned real matrices of the base pass through the same map at the orbit–stabilizer factorization below (Lemma 9.9).

Lemma 9.6 (Homomorphism identities)**.** cayleyProjective *is a group homomorphism, and*

$$M(z+w) = M(z)M(w), \qquad D(z+w) = D(z)D(w).$$

Proof. The homomorphism is the same cancellation that ran the conjugation: for any two invertible matrices M_1, M_2,

$$C\,(M_1 M_2)\,C^{-1} = C\, M_1\, (C^{-1}C)\, M_2\, C^{-1} = (C\, M_1\, C^{-1})(C\, M_2\, C^{-1}).$$

Matrix multiplication with $e^{z+w} = e^z e^w$ gives $M(z+w) = M(z)M(w)$, and applying the homomorphism cayleyProjective to that equation gives $D(z+w) = D(z)D(w)$. □

Every later appearance of cayleyProjective, positioning the distinguished element over the base and acting by a stabilizer factor, is this same map applied to a different matrix. Thus exponentiation is the homomorphism turning addition of logarithmic lifts into composition of the actual slice Möbius maps — recorded once, because every GPV lift below inhabits it:

$$\begin{tikzcd}
\Gamma(t) \arrow[r, "\exp"] \arrow[d, "D"'] & v(t) \in \mathbb{C}^{\times} \arrow[d, "u \mapsto \mathrm{cayleyProjective}(\mathrm{diag}(u,1))"] \\
D(\Gamma(t)) \arrow[r, equal] & \mathrm{cayleyProjective}\big(\mathrm{diag}(v(t),1)\big)
\end{tikzcd}$$

The top row is the logarithmic lift; the bottom row is the Möbius map it generates; the square commutes at every instant, and adding a winding $2\pi ik$ moves only the top row, $e^{\Gamma+2\pi ik} = e^{\Gamma}$. The Euler product will arrive as one completed logarithmic sum, and this square is how one sum acts as one group element at every instant. In the Lean source these are the chain `diagonalGL`, `diagonalGLHom`, `expUnit`, and `diskExpAction`, with their multiplication identities.

Slice preservation is itself a group action, and already one exponentiation carries its GPV degenerate base. For $u \in \mathbb{C}^{\times}$ the logarithms of u form the exponential fibre

$$\exp^{-1}(u) = \{\lambda + 2\pi ik : \ k \in \mathbb{Z}\}, \qquad M(\lambda + 2\pi ik) = M(\lambda),$$

the branches carried by the logarithm manifold of Theorem 6.2 (the commuting triangle $\pi \circ E = \exp$): a lift lives in this fibre, the winding is the integer k separating its elements, and M carries the whole fibre to the one matrix.

The Euler product enters the group through its logarithmic coordinate. There the infinite product is an infinite *sum*, and local normal convergence makes the completed sum a single complex number at every point where it is stated. (*Locally normally* means: every point has a neighbourhood on which the sum of suprema $\sum_p \sup|\ell_p|$ converges. This is the sense in which the cited Euler and Weierstrass products converge, Theorem 1.2 and Proposition 10.1; it gives uniform convergence on compacts, hence continuity of the completed sum, and it is carried in the development as the summability and majorant hypotheses of the A-section interface.) Exponentiation acts once, on the completed sum.

The slice facts of Part 2 make this passage well defined, and they act in order. First the commuting triangle of Theorem 6.2, the GPV logarithm manifold: the $E_{\mathbb{O}}^{+}$-exponential E and the first-factor projection π close over the exponential,

$$\begin{tikzcd}
\mathbb{O} \arrow[r, "E"] \arrow[dr, "\exp"'] & E_{\mathbb{O}}^{+} \arrow[d, "\pi"] \\
& \mathbb{O} \setminus \{0\},
\end{tikzcd} \qquad \pi \circ E = \exp\,.$$

The target is $\mathbb{O} \setminus \{0\}$ because the exponential is nowhere zero (Theorem 6.1); the cited spaces are the uncompactified ones of the literature, read on $\mathbb{O}^*$ as everywhere through Remark 3.6. The blow-up unwraps the multivalued logarithm once and for all: the fibre of π over a value v is the exponential fibre $\exp^{-1}(v)$, its elements the branches, carried into the manifold by E. And the fibre forces one real number. A branch is any λ with $e^{\lambda} = v$; taking norms, the slice exponential gives $\|v\| = e^{\mathrm{Re}\,\lambda}$ (Theorem 6.1), so

$$\mathrm{Re}\,\lambda = \log\|v\|$$

for every branch at once: two branches differ by the purely imaginary $2\pi ik$, and the real coordinate is a function of the value alone. This is the unique GPV real fibre, taken up in detail when the A-section functors are built below. Along a path δ, a continuation of the logarithm is a lift Γ with $e^{\Gamma(t)} = v(t)$; it exists whenever the value stays away from zero along the path, and tameness makes it unique once its initial value $\Gamma(0)$ is fixed in the exponential fibre (Theorem 6.5, Proposition 6.6). The winding is the integer: two lifts of the same value path differ by one constant $2\pi ik$, and exponentiation is blind to it, $e^{\Gamma+2\pi ik} = e^{\Gamma}$, so every branch generates the same matrix at every instant while the real part $\operatorname{Re}\Gamma(t) = \log\|v(t)\|$ is the one real value fixed by the degenerate fibre. This is the general well-definedness: for any slice-preserving section and any pole-avoiding, zero-free path, the completed logarithmic data determines one group element and one real value at each instant, independent of every choice made along the way.

The A-section hypotheses

Here the hypotheses enter — their first working appearance, so we state them informally at once; the formal statement is Definition 10.5. An *A-section* is a section $A \in \mathcal{R}$ with: (C1) a meromorphic continuation with exactly one pole — simple and real — whose pole-cancelled factor continues through that finite point with a finite nonzero value; (C2) an infinite prime-indexed Euler product on a right half-space, each term a power series in p^{-q} for its own prime — one logarithmic base; (C3) an infinite slice-regular Weierstrass factorization whose sphere factors enumerate the residue-$\mathbb{C}$ zeros; (C4) infinitely many such zeros. The hypotheses manufacture one number: C1 continues the pole-cancelled factor through the simple pole, C2 and C3 present that same factor on a punctured neighbourhood of the pole, and its value at the pole is the multiplier $u_A \neq 0$ — the infinite structure completed and evaluated, which M turns into one matrix. No evaluation at the projective north object is part of this analytic continuation.

The general machine now specializes to what the hypotheses supply. Shown at the A-section itself, on the zero-free half-space of C2:

$$\Gamma_A(z) = \sum_p \ell_p(z), \qquad \ell_p(z) = \sum_{k\geq 1} \tfrac{a_{p,k}}{k}\, p^{-kz}, \qquad e^{\Gamma_A(z)} = A(z), \qquad M\big(\Gamma_A(z)\big) = \operatorname{diag}\big(A(z), 1\big) :$$

the infinite prime-power degenerate base completes to one sum, exponentiation acts on it once — $M(z+w) = M(z)M(w)$ is the homomorphism from addition to composition — and the infinite family enters the group as one matrix. In this display and in every later analytic display, z runs on the one compactified plane every slice identifies with, and $A(z)$ is the A-section's one holomorphic stem read there (Definition 6.4): one stem evaluation is the section's evaluation on every slice at once. This is also why the half-space is zero-free: on it $A = e^{\Gamma_A}$ is a value of the exponential, and the exponential is nowhere zero (Theorem 6.1). The unique tame lift re-shows the same construction along any path: a continuation Γ with $e^{\Gamma(t)} = A(\delta(t))$ and fixed initial value is unique (Theorem 6.5), so $M(\Gamma(t))$ is the matrix of the value at every instant, the same for every branch of the fibre.

The slice projection $A^{\text{slice}} : \mathcal{B} \to \text{SphereWorld}$

Conditions C1–C3 construct one element, and this subsection builds it and the slice functor it defines.

Definition 9.7 (The multiplier and the distinguished disk action D_A)**.** Write p_A for the pole of A and $g_A =$ `distinguishedPoleFactor A` for the pole-cancelled factor. On a punctured neighbourhood of the pole, conditions C2 and C3 present this one factor twice:

$$g_A(z) = (z - p_A)\exp\left(\sum_p \ell_p(z)\right),$$
$$g_A(z) = z^{m_A} R_{\mathrm{fac},A}(z) e^{h_A(z)} \prod_{n\geq 0} \mathcal{E}\big(z; q_{A,n}\big).$$

Condition C1 continues this same factor through the simple pole — the finite point p_A, never the domain point ∞ (Remark 1.7); the sum diverges at p_A itself, the factor does not — and evaluates it there:

$$u_A = g_A(p_A) \neq 0.$$

This is the one number the hypotheses manufacture: the infinite structure completed and evaluated. Its logarithms form the exponential fibre of the previous subsection, $\exp^{-1}(u_A) = \{\lambda_A + 2\pi i k : \ k \in \mathbb{Z}\}$, and the function/group-element passage is the commuting square

$$\begin{tikzcd}
z \arrow[r, "\exp"] \arrow[d, mapsto] & e^z \in \mathbb{C}^\times \arrow[d, mapsto] \\
M(z) = \mathrm{diag}(e^z, 1) \arrow[r, "\mathrm{cayleyProjective}"'] & D(z) \in \mathrm{M\ddot{o}b}(\mathbb{C}^*).
\end{tikzcd}$$

through which M collapses the fibre to the one matrix

$$M(\lambda_A) = \mathrm{diag}\big(e^{\lambda_A}, 1\big) = \mathrm{diag}(u_A, 1),$$

so the element the hypotheses construct is

$$D_A = D(\lambda_A) = \mathrm{cayleyProjective}\big(\mathrm{diag}(u_A, 1)\big).$$

Here λ_A is the logarithm and M is what undoes it: the entry of $M(\lambda_A)$ is $e^{\lambda_A} = u_A$, the multiplier itself, so D_A is an exponential action and not a logarithmic one. The logarithm is the coordinate in which the lift moves; the element it generates lives one exponential above it. Every element of the fibre gives this same matrix and hence this same D_A: the logarithm carries the winding, and the exponential action it generates is one element. This is what lets the lift run through the whole base while D_A stays one element. The Euler and Weierstrass expressions displayed above are the two presentations of this same factor.

The element is now fixed. To position it over the projective base, we first construct the projective elements and the unique stabilizer factor that the slice functor uses.

Definition 9.8 (The orbit representatives)**.** For each point b of the base fix one projective element o_b with $o_b \cdot N = b$, once — the development's `GreatCircle.orbitRep` — with $o_N = 1$ at the north object.

Lemma 9.9 (North stabilizer uniqueness)**.** *Every arrow $h : a \to b$ of $\mathcal{B}$ factors through the north stabilizer* $\mathrm{Stab}(N) = \{r : r \cdot N = N\}$ *in exactly one way,*

$$h = o_b\, r_h\, o_a^{-1}, \qquad r_h = o_b^{-1} h\, o_a \in \mathrm{Stab}(N),$$

and the stabilizer factor obeys the identity and composition equalities

$$r_{1_a} = 1, \qquad r_{h_1 ⨾ h_2} = r_{h_2}\, r_{h_1}$$

for every composable pair $h_1 : a \to b$, $h_2 : b \to c$. Groupwise every symbol lives in the one group: h, o_a, o_b, and r_h are elements of $\mathrm{PGL}(2, \mathbb{R})$*, classes of invertible real matrices, and r_h lies in the subgroup* $\mathrm{Stab}(N)$*; the factorization is an equation in* $\mathrm{PGL}(2, \mathbb{R})$*.*

Proof. Take an arbitrary arrow of the base, $h : a \to b$: one element of $\mathrm{PGL}(2, \mathbb{R})$ with $h \cdot a = b$. Its source a has the fixed presentation $o_a \cdot N = a$, its target b has $o_b \cdot N = b$. Form the product $o_b^{-1} h\, o_a$ and let it act on N, one factor at a time, the right factor first:

$$(o_b^{-1} h\, o_a) \cdot N = o_b^{-1} \cdot \big(h \cdot (o_a \cdot N)\big) = o_b^{-1} \cdot (h \cdot a) = o_b^{-1} \cdot b = N.$$

The first equality is the group-action calculation; the second is the defining equation of o_a; the third is $h \cdot a = b$; and the last inverts the defining equation of o_b, from $o_b \cdot N = b$ to $o_b^{-1} \cdot b = N$. So $r_h = o_b^{-1} h\, o_a$ lies in $\mathrm{Stab}(N)$, and $h = o_b\, r_h\, o_a^{-1}$. The factor is unique: if $h = o_b\, r\, o_a^{-1}$ for any stabilizer element r, multiplying on the left by o_b^{-1} and on the right by o_a gives

$$r = o_b^{-1} h\, o_a = r_h.$$

For the identity, the identity arrow $1_a : a \to a$ has source and target a, so the formula $r_h = o_b^{-1} h\, o_a$ reads with o_a on both sides, and the representatives cancel:

$$r_{1_a} = o_a^{-1}\, 1\, o_a = 1.$$

For composition take a composable pair — the codomain of h_1 is the domain of h_2 — $h_1 : a \to b$ and $h_2 : b \to c$. Their composite $h_1 \mathbin{⨟} h_2 : a \to c$ is composition in diagrammatic order, h_1 first, then h_2, and as matrices it is the product $h_2 h_1$. Its ends are a and c, so the formula reads $r_{h_1 ⨟ h_2} = o_c^{-1}(h_2 h_1)\, o_a$, and inserting $o_b\, o_b^{-1} = 1$ between the two matrices splits it into the two stabilizer factors:

$$r_{h_1 ⨟ h_2} = o_c^{-1}(h_2 h_1)\, o_a = (o_c^{-1} h_2\, o_b)(o_b^{-1} h_1\, o_a) = r_{h_2}\, r_{h_1}.$$

□

The chosen transporter from the reference object N to each base point b is therefore the fixed o_b, and what an arbitrary arrow adds is exactly its uniquely determined stabilizer factor r_h. This is a normalization inside the displayed action matrices, not a projection functor and not an analytic evaluation at N.

Definition 9.10 (The slice functor A^{slice})**.** The direction-and-Möbius projection is

$$A^{\mathrm{slice}} : \mathcal{B} \longrightarrow \mathrm{SphereWorld},$$

written `AsectionSlice A` in the Lean source. It is a functor, and its two assignments are built from three constructed pieces: one projective element o_b with $o_b \cdot N = b$, fixed for each object at Definition 9.8, through which every arrow h from a to b factors as $h = o_b\, r_h\, o_a^{-1}$ (Lemma 9.9); the disk action D_A; and cayleyProjective, carrying the base's matrices onto the slice spheres. On objects, $A^{\mathrm{slice}}(b)$ is the normalized slice-sphere object at the already constructed position

$$\mathrm{cayleyProjective}(o_b)\, D_A.$$

On arrows: h goes to the sphere-world arrow whose G_2 automorphism is the identity and whose Möbius transformation is the factorization read through the same builds,

$$\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}.$$

What makes the two assignments well defined is D_A: it is the one factor either formula takes from the A-section, and every branch of the exponential fibre gives that same element (Definition 9.7). Functoriality is proved at Proposition 9.11, read through the disk action: $D(z + w) = D(z)D(w)$ turns sums into compositions, and cayleyProjective carries $r_{1_a} = 1$ and $r_{h_1 ⨟ h_2} = r_{h_2} r_{h_1}$ onto the Möbius transformations (⨟ is composition in diagrammatic order, the left factor first); equal completed data gives equal transformations.

Proposition 9.11 (A^{slice} is a functor)**.** *The two assignments of Definition 9.10 — the positioned disk action at each object and the sphere-world arrow at each arrow — define a functor* $A^{\mathrm{slice}} : \mathcal{B} \to \mathrm{SphereWorld}$*.*

Proof. At an object b, the representative o_b is one element of $\mathrm{PGL}(2, \mathbb{R})$; cayleyProjective carries it onto the slice spheres, and the action at b is

$$\mathrm{cayleyProjective}(o_b)\, D_A.$$

At the north position of the projective base, the chosen representative is the identity, so the action placed in that fibre is D_A. This is a statement about projective positioning; it does not evaluate the A-section at the domain point N. At an arrow $h : a \to b$, the stabilizer factor r_h of Lemma 9.9 gives the Möbius transformation

$$\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}. \tag{OS}$$

Read from right to left, the inverse removes the source position, $\mathrm{cayleyProjective}(r_h)$ applies the factor contributed by h, and the left factor places the result at the target.

For identities, $r_{1_a} = 1$, so the Möbius transformation is

$$\big(\mathrm{cayleyProjective}(o_a)\, D_A\big)\ \mathrm{cayleyProjective}(1)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1} = 1;$$

its G_2 component is also the identity. For $h_1 : a \to b$ and $h_2 : b \to c$, the identity $r_{h_1 \fatsemi h_2} = r_{h_2} r_{h_1}$ and the homomorphism cayleyProjective give

$$\begin{aligned}
&\big(\mathrm{cayleyProjective}(o_c)\, D_A\big)\ \mathrm{cayleyProjective}(r_{h_2})\ \mathrm{cayleyProjective}(r_{h_1})\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1} \\
&\quad = \Big(\big(\mathrm{cayleyProjective}(o_c)\, D_A\big)\ \mathrm{cayleyProjective}(r_{h_2})\ \big(\mathrm{cayleyProjective}(o_b)\, D_A\big)^{-1}\Big) \\
&\qquad \Big(\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_{h_1})\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}\Big).
\end{aligned}$$

The left side is the transformation assigned to $h_1 \fatsemi h_2$, and the right side is the composite of the transformations assigned to h_1 and h_2. Hence A^{slice} preserves identities and composition. $\square$

The orbit–stabilizer calculation has now completed its one role: it made the object and arrow assignments of A^{slice} well defined and proved that they preserve identities and composites (Proposition 9.11). From this point onward the proof uses the resulting functor itself. Its internal factorization is not reopened in the constructions of $\mathcal{A}_A$, $\mathcal{T}_A$, or $\mathcal{R}_A$.

Lemma 9.12 (Normalized pole-fibre transitivity)**.** *For any two normalized coordinates u and v in the same open half-plane, there is a concrete invertible matrix morphism $k : p_A \to p_A$ in $\mathcal{B}$ such that*

$$A^{\mathrm{slice}}(p_A)^{-1} \cdot \big(A^{\mathrm{slice}}(k) \cdot (A^{\mathrm{slice}}(p_A) \cdot u)\big) = v. \tag{PT}$$

Proof. The concrete real projective matrix in (PGL), with the base source and target both p_A, fixes p_A and carries any prescribed point of the open half-plane to any other. Reading that calculation in the normalized coordinate at p_A conjugates its matrix action by the fixed invertible action $A^{\mathrm{slice}}(p_A)$. Conjugation preserves products, inverses, and transitivity, and gives exactly the left side of (PT). The resulting matrix is the stated morphism $k : p_A \to p_A$. $\square$

A^{slice} records the spheres and the positioned Möbius motion, and the equivariant realization $A_{\mathbb{O}}$ below realizes the section's value on them. The full A-section functor below retains, in addition, the normalized point and the value produced on that sphere.

Exponentiating the prime-power sum — the A-section action whose pole-cancelled continuation has the C3 Weierstrass presentation at the finite point p_A — is why there is a fixed real-value transport: the path from $\delta(0)$ to $\delta(1)$ determines the family of matrix actions $D(\Gamma(t))$, together with its real level at every instant. The GPV transport of the next subsection records exactly this.

The GPV transport

The GPV lift records how g_A varies between its presentations.

Definition 9.13 (The GPV transport)**.** Let $a, b \in \mathcal{B}$. By the I-independent lift (Definition 6.4), evaluating the A-section off the real axis is evaluating its one holomorphic stem on the one compactified plane every slice identifies with; a projective-base transport from a to b records that stem evaluation along a path. It carries three synchronized data, each a continuous function of one parameter $t \in [0, 1]$, the instant of the transport: a domain path δ^* — at each instant t, the point of that plane at which the stem is evaluated, running from the point of a to the point of b; the value v the section takes there; and a logarithmic lift Γ, the continuous logarithm of that value. The instant parametrizes the transport from a to b and belongs to the transport alone; the geometry keeps its own coordinates, and the three data are read together at each instant. The hypotheses are what make the instant necessary. C1 is a continuation, and continuation happens along a path: the one factor is continued to the pole, and its value at each instant is the continued value. The logarithm of C2's completed sum becomes single-valued along a path: the unique tame lift fixes every branch from the initial one, which is the GPV mechanism itself. And the argument reads what the path carries: the one pole-cancelled factor from its Euler presentation on the half-space to its Weierstrass presentation at the finite pole p_A, one real value carried the whole way. What varies with the instant is the prime-power sum: the degenerate base of C2 is evaluated along the domain path, and the matrix it generates varies with it. The matrix acts throughout on one fixed input point of its locus. The beginning and arrival of the path are determined by the hypotheses: it begins in the Euler half-space of C2, where the prime-power degenerate base converges, and ends at the pole of C1; the instants $t = 0$ and $t = 1$ are this beginning and this arrival. Here the critical strip's classical mystery takes its present form: the strip lies beyond the half-space where the prime-power base converges, and the zeros stand there as fixed coordinates; the path is how the arithmetic of the primes reaches them, as the matrix produced at the pole. The value at the pole generates that matrix, and the matrix acts on the standing input: the prime-power sum varies in the construction of the matrix, and the input it acts on is fixed. The section takes its values on the compactified sphere, and the exponential fibres live one exponential below, over $\mathbb{C}^\times$; so the value is carried twice at once, as the finite nonzero $v(t) \in \mathbb{C}^\times$ that the exponential reaches, and as its reading $v^*(t)$ on the compactified sphere, the same value seen where the section takes it. Their defining equations are

$$\begin{aligned} \delta^*(0) &= \mathrm{complexPoint}(a), & \delta^*(1) &= \mathrm{complexPoint}(b), \\ v^*(t) &= A\big(\delta^*(t)\big), & e^{\Gamma(t)} &= v(t), \\ \Gamma(1) - \Gamma(0) &= 2\pi i k, & k &\in \mathbb{Z} \quad \text{when the value path is a circuit :} \end{aligned}$$

the domain path runs from a to b; the value, read on the sphere, is the section's value along it; and the lift exponentiates to the value where the exponential lives. The last equation is not asserted for a general path: it records the circuit case, when the source and target values agree, and k is its winding integer. These are precisely the fields of `ASection.GpvTransport`.

At each t the lift equation passes through the matrix square:

$$\begin{tikzcd}
\Gamma(t) \arrow[rr, "\exp"] \arrow[d, "D"'] & & v(t) \in \mathbb{C}^\times \arrow[d, "u \mapsto \mathrm{cayleyProjective}(\mathrm{diag}(u,1))"] \\
D(\Gamma(t)) \arrow[r, equal] & \mathrm{cayleyProjective}\big(\mathrm{diag}(v(t),1)\big). &
\end{tikzcd}$$

Thus every point of the lift is already an action of the same kind as D_A.

Lemma 9.14 (The real level)**.** *Along a GPV transport the real level* $\operatorname{Re}\Gamma(t) = \log\|v(t)\|$ *is continuous, depends only on the value along the path, and is the same for every lift of that value path.*

Proof. The lift equation is $e^{\Gamma(t)} = v(t)$. Take norms: the slice exponential has $\|e^{\Gamma(t)}\| = e^{\operatorname{Re}\Gamma(t)}$ (Theorem 6.1), so

$$e^{\operatorname{Re}\Gamma(t)} = \|v(t)\|, \qquad \operatorname{Re}\Gamma(t) = \log\|v(t)\| :$$

the real level is continuous and depends only on the value along the path. Uniqueness and branch-independence are the cited GPV facts (Theorem 6.5, Proposition 6.6): the tame lift is unique once its initial value $\Gamma(0)$ is fixed, and any two lifts of the same value path differ by one purely imaginary constant $2\pi i k$, so every lift carries the same real part ([21, Cor. 4.13]). □

At each instant the action is $D(\Gamma(t))$, moving with the value along the domain path: one group element and one real value at every instant, the branch ambiguity spent fibrewise on the degenerate fibre. The winding field records the circuit case, the path whose value returns, and its integer counts the branches crossed. This is the fixed real-value transport: the path from $\delta(0)$ to $\delta(1)$ determines the matrix actions, and their real levels define the transported real value.

Condition C2 now supplies the canonical arithmetic instance of this construction. Along an Euler half-space loop δ,

$$\Gamma_A(t) = \sum_p \ell_p(\delta(t)), \qquad e^{\Gamma_A(t)} = A(\delta(t)),$$

with the prime index retained inside the complete summation. Local normal convergence makes Γ_A continuous, the zero-free half-space makes its value lie in $\mathbb{C}^\times$, and the GPV uniqueness theorem determines the whole lift from its initial value $\Gamma_A(0)$ in the exponential fibre. The sum is complete at every instant; only the evaluation point moves along the domain path. Since $\delta(0) = \delta(1)$, the completed sum is evaluated at the same point at both ends, so the checked winding is $k = 0$: the loop returns to the same element of the fibre, $\Gamma_A(1) = \Gamma_A(0)$, level and matrix with it.

The same construction continues to the finite pole. This analytic passage uses the finite continued multiplier and its exponential fibre; it does not introduce a new projective object or a new value of the A-section.

Lemma 9.15 (Finite-pole arrival)**.** *Run the domain path of a GPV transport of the continued factor to* $\delta(1) = p_A$*. Then*

$$v(1) = g_A(p_A) = u_A, \qquad D(\Gamma(1)) = D(\lambda_A) = D_A, \qquad \operatorname{Re}\Gamma(1) = \operatorname{Re}\lambda_A,$$

and two such runs differ by one whole winding, $\Gamma_2(1) - \Gamma_1(1) = 2\pi i k$ *with* $k \in \mathbb{Z}$*.*

Proof. The value along the run is $g_A(\delta(t))$: condition C1 continues g_A through the simple pole, so the value is nonzero the whole way, entering from the Euler half-space where $g_A = (z - p_A)\, e^{\Gamma_A}$ (condition C2) and arriving at $v(1) = g_A(p_A) = u_A$, where condition C3 presents the multiplier u_A in Weierstrass form. The arriving action is D_A as Definition 9.7 constructed it, $D_A = D(\lambda_A) = \mathrm{cayleyProjective}\big(\mathrm{diag}(u_A, 1)\big)$, and its level is $\operatorname{Re}\lambda_A$ by Lemma 9.14. Every finite-pole run therefore produces this same matrix action and the same real value. The two lifts differ by one purely imaginary constant $2\pi i k$, the whole winding; exponentiation removes that difference, while taking real parts removes it as well. □

The prime sum, its value under A, its logarithmic level and winding, and the matrix action it generates are the data of one element, $D(\Gamma(t))$ at each instant, travelling together.

Condition C4, infinitely many residue-$\mathbb{C}$ zeros, matches the class to its infinite structure — the infinite Euler family of C2, the infinite Weierstrass divisor of C3 — and populates the degenerate fibre of the transport with an infinite family; Theorem 10.9 reads one real value across the whole population.

The equivariant realization $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$

An A-section is a ring element whose value at each point belongs to that point's own slice sphere. The slice functor just built records the positioned sphere-world action. The total Grothendieck construction also uses its point-valued realization: input in $\mathbb{O}^*$, output in $\mathbb{O}^*$, and the same G_2 element on each equivariant arrow. This subsection defines that functor.

Definition 9.16 (The equivariant realization)**.** The realization is a functor between two groupoids already built, from the octonionic world to itself:

$$A_{\mathbb{O}} : \mathcal{H}_1 \longrightarrow \mathcal{H}_1, \qquad \mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*,$$

written `AsectionEquivariant` in the Lean source. It is the point-valued realization of the same slice-preserving action whose positioned sphere-world projection is A^{slice}. On an object $q = x + Iy$ it uses the stem already constructed in Definition 9.2:

$$A_{\mathbb{O}}(x + Iy) = F_1(x + iy) + I\, F_2(x + iy),$$

with the compactified value prescribed at N. Thus no second section or independent value map is introduced. On arrows, an arrow of $\mathcal{H}_1$ is one element $g \in G_2$ with $g \cdot q = q'$, and $A_{\mathbb{O}}$ sends it to that same element,

$$A_{\mathbb{O}}\big(g : q \to q'\big) \;=\; g : A_{\mathbb{O}}(q) \to A_{\mathbb{O}}(q').$$

Proposition 9.17 ($A_{\mathbb{O}}$ is a functor)**.** *The two assignments of Definition 9.16 define a functor $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$.*

Proof. The target is an arrow of $\mathcal{H}_1$ because equivariance (Definition 9.2(ii)) gives $g \cdot A_{\mathbb{O}}(q) = A_{\mathbb{O}}(g \cdot q) = A_{\mathbb{O}}(q')$. Preservation of identities and composition is the group's own: the identity $1 : q \to q$ goes to $1 : A_{\mathbb{O}}(q) \to A_{\mathbb{O}}(q)$, and a composite hg goes to hg, the same group elements on both sides. □

A^{slice} and $A_{\mathbb{O}}$ are two functors of the same slice-preserving action: the first records the projective direction-and-Möbius transport, the second the value of the section, retaining the same G_2 arrow by equivariance. Condition C2 identifies the element acting in both with the exponentiated log sum of prime powers — the Euler product — of the given A-section; after the continuation of C1 through the finite pole p_A, condition C3 presents the same pole-cancelled multiplier in Weierstrass form and selects the residue-$\mathbb{C}$ locus; and C4 makes the selected population infinite. The A-section functor of the next subsection uses both.

The unique GPV real fibre

The projective group theory and the functoriality of A^{slice} are already proved. The remaining analytic fact is the branch-independence supplied by the GPV lift.

Remark 9.18 (The unique GPV real fibre). The real fibre is what is exponentiated. Over each base object, the branches of the prime-power lift differ by $2\pi ik$. Consequently they have the same real part, and M sends every branch to the same matrix:

$$\mathrm{Re}(\lambda + 2\pi ik) = \mathrm{Re}\,\lambda, \qquad M(\lambda + 2\pi ik) = M(\lambda)$$

(Definition 9.7). Local normal convergence makes the completed prime-power sum one complex number at each point where it is defined; its exponential is one value, and its logarithms form one exponential fibre. At the pole p_A, condition C3 presents the completed multiplier u_A in Weierstrass form, and

$$\exp^{-1}(u_A) = \{\lambda_A + 2\pi ik : \ k \in \mathbb{Z}\}$$

has the single real level $\mathrm{Re}\,\lambda_A$.

The analytic datum used below is the exponential fibre over the continued multiplier: its branches differ by $2\pi ik$, every branch has the same real part by GPV uniqueness, and every branch exponentiates to the same A-specific disk action.

The functor for the total Grothendieck construction: $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$

The total Grothendieck construction of the next subsection takes one input: a functor from the base to **Grpd**, and every ingredient of this one is already built. An object of **Grpd** is a groupoid; $\mathcal{H}_1$, the sphere world, and $\mathcal{B}$ are three of its objects. An arrow of **Grpd** is a functor between groupoids, and the two functors just built are two of its arrows: $A^{\mathrm{slice}} : \mathcal{B} \to \mathrm{SphereWorld}$ records where the inputs are positioned, and $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$ records the values A^{slice} produces. The functor this subsection builds lands in **Grpd** itself:

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd}.$$

To each base object b it assigns a groupoid: its objects are the A-section's states at b, and its arrows are elements of the group G_2, one element γ with $\gamma \cdot x = y$ — G_2 acting on those states. To each projective element $h : a \to b$ it assigns the transport of those states: each state moves by the concrete Möbius matrix action of $A^{\mathrm{slice}}(h)$ (`projectiveArrowHom`). The states are positioned by the A-section's action, and the hypotheses built that action once, at the pole p_A: C1 continues g_A through the pole, the value u_A there generates the disk action D_A, and C3 gives it its Weierstrass form. The total construction glues these groupoids, one per point of the real great circle, into $\mathcal{T}_A$ of the next subsection. The source $\mathcal{B} = \mathrm{PGL}(2,\mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$ is the projective action groupoid: an object is a point of the compactified real projective line, and an arrow $h : a \to b$ is one element of $\mathrm{PGL}(2,\mathbb{R})$ with $h \cdot a = b$. The base is this circle for the A-section's own reason: the compactified real axis is the one line every slice shares, the finite pole p_A is one of its objects, and every base arrow is an invertible projective matrix whose cayleyProjective action is an invertible Möbius transformation on the corresponding slice spheres. The A-specific disk action these matrices transport is the exponential of the complete prime-power GPV lift, determined by its initial value in the exponential fibre.

Each point of $\mathbb{O}^*$ off the real axis lies on exactly one slice sphere, so it is two data: the sphere, named by its direction $I \in S^6$, and the point on it, $q = x + Iy$. Its slice coordinate is the complex number $u = x + iy$, the point read on the one compactified plane $\mathbb{C}^*$ that every slice identifies with — normalized means read there, the same plane for every direction. The pair (I, u) is that point in components, $q = (I, u)$, paired as the normalized value states introduced with the sphere world, and $\gamma \cdot (I, u) = (\gamma I, u)$.

Definition 9.19 (The A-section states and their fibre groupoids)**.** On a base object b, the groupoid $\mathcal{A}_A(b)$ consists of the A-section states positioned at b by A^{slice}. A state is one triple,

$$\Big((I, u), \quad A^{\text{slice}}(b) \cdot (I, u), \quad A_{\mathbb{O}}\big(A^{\text{slice}}(b) \cdot (I, u)\big) \Big) :$$

a slice direction $I \in S^6$ with a normalized input coordinate u, the coordinate positioned at b, and the value produced there. Only the input is free. The positioned entry is determined by the position, and the value entry is the value $A_{\mathbb{O}}$ computes at the positioned entry: the second and third entries are one point of the equivariant realization's graph. The middle entry is a coordinate on its slice Riemann sphere and the third entry lies in $\mathbb{O}^*$. The label $b = p_A$ means that the triple lies in the fibre positioned at the finite pole; it does not say that its middle entry is the point p_A, and therefore does not force its value entry to be N. In particular, a residue state in that fibre has a C-residue zero in its middle entry and zero in its value entry. The action used here is constructed from the finite value $u_A = g_A(p_A) \neq 0$ of the pole-cancelled factor that condition C1 continues through the pole.

For two such triples x and y, define

$$\mathrm{Hom}_{\mathcal{A}_A(b)}(x, y) := \{\gamma \in G_2 : \gamma \cdot x = y\}.$$

Thus an arrow is a single element $\gamma \in G_2$ acting on the slice direction and carrying the rest of the triple with it. The disk action, conjugated by cayleyProjective, is the slice-preserving action on each Riemann sphere of the continuum in SphereWorld, the same complex map on every sphere: in components $D \cdot (I, u) = (I,\ D \cdot u)$, while γ moves the direction, $\gamma \cdot (I, u) = (\gamma I,\ u)$. Each acts on its own component, so the two commute:

$$D \cdot (\gamma \cdot q) = D \cdot (\gamma I,\ u) = (\gamma I,\ D \cdot u) = \gamma \cdot (I,\ D \cdot u) = \gamma \cdot (D \cdot q) :$$

the positioned entry moves coherently, and the value entry moves by equivariance (Definition 9.2(ii)). The identity arrow of x is $1 \in G_2$; if $\gamma : x \to y$ and $\delta : y \to z$, their composite is $\delta\gamma : x \to z$ because

$$(\delta\gamma) \cdot x = \delta \cdot (\gamma \cdot x) = z;$$

and $\gamma^{-1} : y \to x$ is the inverse arrow. Associativity, the identity equalities, and the two inverse equalities are those of the group G_2. Consequently $\mathcal{A}_A(b)$ is, concretely, the action groupoid of G_2 on the displayed triples; in particular every arrow is invertible. The fibres assemble the equivariant realization's graph, and their value entries are where condition C3 selects the residue-$\mathbb{C}$ zero states the concentricity theorem reads. In the Lean source the pairs are the states `AsectionState A`, whose point and value maps into $\mathcal{H}_1$ are `AsectionStateInput A` and `AsectionStateOutput A`, natural transformations checked with `AsectionState_input_then_equivariant`; the triples are `AsectionActionState`, their G_2 action groupoid is `AsectionActionStateFiber`, and `AsectionActionFiber A X` is the fibre groupoid of the displayed triples at the base object X.

Definition 9.20 (The transport)**.** On a base arrow $h : a \to b$, $\mathcal{A}_A(h)$ is a functor between the two fibre groupoids,

$$\mathcal{A}_A(h) : \mathcal{A}_A(a) \longrightarrow \mathcal{A}_A(b),$$

called *the transport*. The arrow h is already a concrete invertible matrix morphism of the groupoid $\mathcal{B}$, and $A^{\text{slice}}(h)$ is already its concrete invertible matrix action in SphereWorld. Definition 9.20 uses that functorial action directly; it does not reopen the construction that proved A^{slice} well defined.

The middle entry of a state at a is $A^{\text{slice}}(a) \cdot (I, u)$. Apply the matrix action $A^{\text{slice}}(h)$ to that entry. The normalized input of the resulting state at b is obtained by the inverse of the invertible action at b. Thus the whole assignment on a triple is

$$\begin{aligned}
&\Big((I, u),\quad A^{\text{slice}}(a) \cdot (I, u),\quad A_{\mathbb{O}}\big(A^{\text{slice}}(a) \cdot (I, u)\big) \Big) \\
&\quad \longmapsto \Big(\big(I,\ A^{\text{slice}}(b)^{-1} \cdot \big(A^{\text{slice}}(h) \cdot (A^{\text{slice}}(a) \cdot u)\big)\big), \\
&\qquad\quad A^{\text{slice}}(h) \cdot \big(A^{\text{slice}}(a) \cdot (I, u)\big), \\
&\qquad\quad A_{\mathbb{O}}\Big(A^{\text{slice}}(h) \cdot \big(A^{\text{slice}}(a) \cdot (I, u)\big)\Big) \Big).
\end{aligned}$$

This is a state at b because applying the action at b to its displayed normalized input gives its displayed middle entry:

$$\begin{aligned}
&A^{\text{slice}}(b) \cdot \Big(I, A^{\text{slice}}(b)^{-1} \cdot \big(A^{\text{slice}}(h) \cdot (A^{\text{slice}}(a) \cdot u)\big)\Big) \\
&\quad = A^{\text{slice}}(h) \cdot \big(A^{\text{slice}}(a) \cdot (I, u)\big).
\end{aligned}$$

The third entry is then exactly what $A_{\mathbb{O}}$ computes at that middle entry. On a fibre arrow γ, the transport uses the same matrix $\gamma \in G_2$: the projective/Möbius matrices change the complex coordinate and the G_2 matrix changes the slice direction, so their actions commute componentwise. These are the object and arrow assignments (`AsectionActionTransport_obj_input`, `AsectionActionTransport_obj_positioned`, `AsectionActionTransport_obj_value`).

Proposition 9.21 ($\mathcal{A}_A$ is a functor)**.** *The assignments $b \mapsto \mathcal{A}_A(b)$ and $h \mapsto \mathcal{A}_A(h)$ define a functor $\mathcal{A}_A : \mathcal{B} \to$ **Grpd**, written `AsectionActionDiagramA` in the Lean source.*

Proof. Fix a base arrow $h : a \to b$. A fibre arrow is a matrix $\gamma \in G_2$ satisfying $\gamma \cdot (I, u) = (\gamma I, u)$, whereas every matrix occurring in the object assignment of Definition 9.20 acts on the complex coordinate and leaves I fixed. Hence applying γ before that displayed assignment and applying the same γ afterward give the same complete triple. The middle entries agree by the matrix action on the complex coordinate, and the third entries agree by the equivariance of $A_{\mathbb{O}}$. Therefore $\mathcal{A}_A(h)$ sends the fibre arrow γ to that same matrix γ between the transported states. In particular it sends the identity matrix to the identity matrix and matrix products to the same products:

$$\mathcal{A}_A(h)(1_x) = 1_{\mathcal{A}_A(h)(x)}, \qquad \mathcal{A}_A(h)(\delta\gamma) = \mathcal{A}_A(h)(\delta)\, \mathcal{A}_A(h)(\gamma).$$

Thus every $\mathcal{A}_A(h)$ is a functor between the two action groupoids.

For the identity base arrow 1_a, Proposition 9.11 gives $A^{\text{slice}}(1_a) = 1$. The normalized-input entry of Definition 9.20 is therefore

$$A^{\text{slice}}(a)^{-1} \cdot \big(A^{\text{slice}}(1_a) \cdot (A^{\text{slice}}(a) \cdot u)\big) = u.$$

The middle and value entries are likewise unchanged, and the arrow map retains the same G_2 matrix. Hence

$$\mathcal{A}_A(1_a) = 1_{\mathcal{A}_A(a)}.$$

For $h_1 : a \to b$ followed by $h_2 : b \to c$, apply the displayed object assignment twice. On the normalized input the two actions at the intermediate object cancel, and functoriality of A^{slice} replaces the two consecutive matrix actions by the action of the base composite:

$$\begin{aligned}
&A^{\text{slice}}(c)^{-1} \cdot A^{\text{slice}}(h_2) \cdot A^{\text{slice}}(b) \cdot \left(A^{\text{slice}}(b)^{-1} \cdot A^{\text{slice}}(h_1) \cdot A^{\text{slice}}(a) \cdot u\right) \\
&\quad = A^{\text{slice}}(c)^{-1} \cdot A^{\text{slice}}(h_2) \cdot A^{\text{slice}}(h_1) \cdot A^{\text{slice}}(a) \cdot u \\
&\quad = A^{\text{slice}}(c)^{-1} \cdot A^{\text{slice}}(h_1 \mathbin{⨟} h_2) \cdot A^{\text{slice}}(a) \cdot u.
\end{aligned}$$

The same equality gives the positioned entry; applying $A_{\mathbb{O}}$ gives the value entry; and on arrows both sides retain the same G_2 matrix. Therefore

$$\mathcal{A}_A(h_1 \mathbin{⨟} h_2) = \mathcal{A}_A(h_1) \mathbin{⨟} \mathcal{A}_A(h_2).$$

These are exactly `AsectionActionTransport_id` and `AsectionActionTransport_comp`. Hence the assignments define the functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. $\square$

$\mathcal{A}_A$ is the graph of A from the opening of the section, made categorical: input, position, value, the three columns of the picture, with the transport squares as the symmetries that carry graph points to graph points.

The total $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$

The Grothendieck construction gathers the indexed family built above — over each point b of the projective base the fibre $\mathcal{A}_A(b)$ — into a single category, and the construction needs four things of its input. First, the base is a groupoid. Second, the functor assigns a groupoid to each base object. Third, it assigns a functor between those groupoids to each base arrow. Fourth, the assignments are functorial, identities to identities and composites to composites. The first is already established: the base $\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \text{OnePoint}(\mathbb{R})$ is a groupoid because its arrows are group elements — an arrow is one projective element h with $h \cdot a = b$, composition is the group's multiplication, and the group inverse inverts every arrow. The remaining three are met by what the previous subsection built. A groupoid is assigned to each base object: the fibre $\mathcal{A}_A(b)$, the G_2 groupoid of the state triples, each triple constructed from the two functors A^{slice} and $A_{\mathbb{O}}$ (Definition 9.19). A functor is assigned to each base arrow: the transport $\mathcal{A}_A(h) : \mathcal{A}_A(a) \to \mathcal{A}_A(b)$ (Definition 9.20). And the assignments are functorial (Proposition 9.21). With all four in hand, $\mathcal{T}_A$ is a well-defined total groupoid, and this subsection describes its objects and morphisms and proves the groupoid structure.

An object of the total $\int_{\mathcal{B}} \mathcal{A}_A$ is a pair (b, x): a point b of the real great circle and a state x of the fibre $\mathcal{A}_A(b)$ (Definition 9.19), the triple

$$x = \Big((I, u), \quad A^{\text{slice}}(b) \cdot (I, u), \quad A_{\mathbb{O}}\big(A^{\text{slice}}(b) \cdot (I, u)\big) \Big)$$

positioned at b: one graph point of the A-section. The pairing carries analytic content, and it is how this construction was first conceived: by GPV uniqueness, the state over b has one exponential fibre — every branch of the lift over it differs by one purely imaginary constant $2\pi i k$, so all its branches have the same real part, the level the transport carries, and the exponential action every branch generates is one element (Theorem 6.5, Lemma 6.3).

The morphisms of the total are fixed by the transport (Definition 9.20): for two objects (a, x) and (b, x'),

$$\mathrm{Hom}_{\mathcal{T}_A}\big((a,x),(b,x')\big) = \big\{(h,\gamma)\ : h : a \to b \text{ in } \mathcal{B},\\ \gamma : \mathcal{A}_A(h)(x) \to x' \text{ in } \mathcal{A}_A(b)\big\}.$$

Here the base component h is read through $\mathcal{A}_A(h)$, whose transport is the A-specific Euler–Weierstrass action and GPV lift already constructed for h. The pair (h,γ) therefore records the concrete projective matrix action and the displayed G_2 comparison in the target fibre; the GPV data are already part of the functor applied to h, not an additional component of the pair. The states x and x' are triples in the fibres at a and b, respectively. The transport first produces $\mathcal{A}_A(h)(x)$ in the fibre at b; the element γ then compares that triple with x'. Thus a morphism has two stages, base then fibre: h carries the state to the fibre at b, and γ compares directions there, the first entry (I,u) going to $(\gamma I,u)$ with the positioned and value entries moving by the commutation and equivariance displayed at $\mathcal{A}_A$.

Both components are concrete matrix group actions in their corresponding groupoid categories. The projective matrix h satisfies $h \cdot a = b$, and the matrix $\gamma \in G_2$ satisfies $\gamma \cdot \mathcal{A}_A(h)(x) = x'$. Their identity matrices, products, and inverse matrices are the identity morphisms, composites, and inverse morphisms used below.

That these sets are the hom-sets of a groupoid — identities, composites, inverses — is Proposition 9.22 below.

Proposition 9.22 ($\mathcal{T}_A$ is a groupoid)**.** *$\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ is a groupoid, with composite*

$$(h_1,\gamma) \mathbin{⨟} (h_2,\gamma') = \big(h_1 \mathbin{⨟} h_2,\ \gamma'\gamma\big),$$

where $h_1 \mathbin{⨟} h_2$ is multiplication in $\mathrm{PGL}(2,\mathbb{R})$, $\gamma'\gamma$ is multiplication in G_2, and the A-specific slice transports associated to h_1 and h_2 multiply in $\mathrm{Möb}(\mathbb{C}^*) = \mathrm{PGL}(2,\mathbb{C})$. This Möbius transformation is the one already assigned by $A^{\mathrm{slice}}(h_1 \mathbin{⨟} h_2)$, and Proposition 9.21 identifies it with the composite of the two transports. Every morphism has the inverse $(h,\gamma)^{-1} = (h^{-1},\gamma^{-1})$; the GPV transport already belonging to $\mathcal{A}_A(h^{-1})$ is the reversal of the transport belonging to $\mathcal{A}_A(h)$.

Proof. The identity morphism of (a,x) is the pair $(1_a,1_x)$: $\mathcal{A}_A(1_a)$ is the identity functor of the fibre (Proposition 9.21), so 1_x is a fibre arrow from $\mathcal{A}_A(1_a)(x) = x$ to x (`AsectionActionTransport_id`).

For composition take $(h_1,\gamma) : (a,x) \to (b,x')$ followed by $(h_2,\gamma') : (b,x') \to (c,x'')$: so γ is an arrow of the fibre at b with $\gamma \cdot \mathcal{A}_A(h_1)(x) = x'$, and γ' one of the fibre at c with $\gamma' \cdot \mathcal{A}_A(h_2)(x') = x''$. The composite must be a pair over $h_1 \mathbin{⨟} h_2$: one arrow of the fibre at c from $\mathcal{A}_A(h_1 \mathbin{⨟} h_2)(x)$ to x''. It is built in three steps. First, by the composites-to-composites functoriality of Proposition 9.21,

$$\mathcal{A}_A(h_1 \mathbin{⨟} h_2)(x) = \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x)\big).$$

Second, apply the functor $\mathcal{A}_A(h_2)$ to the arrow γ of the fibre at b: by the fibre-arrow computation in the proof of Proposition 9.21, the result is the same group element, now an arrow of the fibre at c,

$$\gamma \cdot \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x)\big) = \mathcal{A}_A(h_2)(x').$$

Third, compose with γ' in the fibre at c — multiplication in G_2:

$$(\gamma'\gamma) \cdot \mathcal{A}_A(h_1 \mathbin{⨟} h_2)(x) = \gamma' \cdot \mathcal{A}_A(h_2)(x') = x''.$$

So $(h_1,\gamma) \mathbin{⨟} (h_2,\gamma') = (h_1 \mathbin{⨟} h_2,\ \gamma'\gamma)$: the base component is multiplied in $\mathrm{PGL}(2,\mathbb{R})$; its associated slice transformation is multiplied in $\mathrm{Möb}(\mathbb{C}^*) = \mathrm{PGL}(2,\mathbb{C})$; and the fibre component is multiplied in G_2. The GPV transport belonging to $\mathcal{A}_A(h_1 \mathbin{⨟} h_2)$ is the composite of the transports already belonging to $\mathcal{A}_A(h_1)$ and $\mathcal{A}_A(h_2)$. Associativity follows from associativity in these groups together with the functoriality that binds the slice Möbius transport to the base composite (`AsectionActionTransport_comp`).

For the inverse of (h, γ): the base is a groupoid, so $h \fatsemi h^{-1} = 1_a$ and $h^{-1} \fatsemi h = 1_b$. Applying the identity and composition equalities of Proposition 9.21 gives

$$\mathcal{A}_A(h) \fatsemi \mathcal{A}_A(h^{-1}) = \mathcal{A}_A(h \fatsemi h^{-1}) = \mathcal{A}_A(1_a) = 1_{\mathcal{A}_A(a)},$$
$$\mathcal{A}_A(h^{-1}) \fatsemi \mathcal{A}_A(h) = \mathcal{A}_A(h^{-1} \fatsemi h) = \mathcal{A}_A(1_b) = 1_{\mathcal{A}_A(b)}.$$

Thus every transport $\mathcal{A}_A(h)$ is an isomorphism in **Grpd** with the two-sided inverse functor $\mathcal{A}_A(h^{-1})$, an automorphism of the fibre $\mathcal{A}_A(a)$ when $b = a$ — and both functors carry each G_2 matrix to that same matrix. Consequently $(h, \gamma)^{-1} = (h^{-1}, \gamma^{-1})$ composes with (h, γ) to the identity pairs on both sides. The total is a groupoid. In the Lean source the construction is Mathlib's `CategoryTheory.Grothendieck` applied to `AsectionActionDiagram A`, and the groupoid classification is `grothendieckGrpdGroupoid`. □

Every fibre consists of the complete A-section triples at its base object; every transport is the concrete matrix action already assigned by A^{slice}; and every value entry is the equivariant realization computed at the transported positioned point. The total is the A-section's graph made one category: what moves from fibre to fibre is the A-specific Euler–Weierstrass/GPV action, and the residue-$\mathbb{C}$ zero-sphere states are the degenerate-fibre outputs of this completed construction, produced by C3–C4. The total is read in two ways, and both are functors it comes with. The projection to the base, $(b, x) \mapsto b$ and $(h, \gamma) \mapsto h$, forgets the state (`CategoryTheory.Grothendieck.forget`); over it the total holds one fibre per point of the real great circle. This projection never moves a state into the pole fibre: passage between fibres, including passage to or from p_A, is always an actual base morphism acted on by the A-specific transport functor. And the components diagram $\pi_0 \circ \mathcal{A}_A : \mathcal{B} \to \mathbf{Set}$ sends each base object to the set of components of its fibre; its colimit enters the readout through Lemma 10.3 — π_0 of the total is the colimit of the fibrewise components, which is how the next section turns connectedness into one real number.

Definition 9.23 (The projective base, the A-section functor, and the total category)**.** The base is the projective action groupoid $\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$. The A-section functor is the assignment constructed in this section — on objects the G_2 action groupoid of state triples positioned at each base object, on arrows transport by the Möbius transformation assigned by the already-proved functor A^{slice} and the A-specific GPV transport — written `AsectionActionDiagram A`, $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. The total value-transport category is $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$.

10 The concentricity theorem

We now have defined our groupoids and functors, so this section defines the class of functions precisely and proves the theorem. Two facts about slice-preserving functions come first: the Weierstrass factorization that gives condition C3 its meaning, and the fact that a residue-$\mathbb{C}$ zero is a whole 6-sphere, a single connected component of the octonionic world. Then the categorical engine, Riehl's computation of the components of a Grothendieck construction as a colimit, which sorts the assembled total into the classes linked by the section's real-value transports. With these in hand we define an A-section by its four conditions C1–C4, state and prove the Concentricity Theorem, and record the corollary that carries its conclusion back into the language of classical zeros.

Proposition 10.1 (Slice-regular Weierstrass factorization; the content of C3)**.** *Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as*

$$A = q^m \, R \, S \, h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order-m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A, S is a factor vanishing exactly at its spherical (residue-$\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds including the case where either family of zeros is infinite.

Proof. The proposition is general: A here is any member of $\mathcal{R}$, the hypotheses C1–C4 play no role in the argument, and either family of zeros may be finite, infinite, or empty. Every analytic step happens one algebra down, at the stem; the slice formalism then lifts the finished factorization to $\mathbb{O}^*$.

The reduction to the stem. By the I-independent lift (Definition 6.4), A is induced by one holomorphic stem $F = F_1 + \iota F_2$ with F_1, F_2 $\mathbb{R}$-valued:

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad \text{the same components for every } I.$$

Real components give $F(\bar{z}) = \overline{F(z)}$, so the zero set of F is conjugation-symmetric: the real zeros ρ_k stand individually, and the non-real zeros stand in conjugate pairs $\{\rho_j, \bar{\rho}_j\}$. Under the lift a real zero is a residue-$\mathbb{R}$ zero of A, and one conjugate pair is one zero-sphere, its whole G_2-orbit (Lemma 10.2).

The classical factorization, symmetrized. A holomorphic function g has real Taylor coefficients exactly when it satisfies the conjugation symmetry $\overline{g(\bar{z})} = g(z)$; this is the form in which realness is tracked below, and it is the symmetry the stem already has. The classical Weierstrass theorem factors F over its zero set: with the primary factors

$$E_p(w) = (1 - w) \exp\Big(w + \tfrac{w^2}{2} + \cdots + \tfrac{w^p}{p}\Big)$$

and the integers p_n chosen so that $\sum_n (r/|\rho_n|)^{p_n+1}$ converges for every $r > 0$, the product converges normally on compact sets and

$$F(z) = z^m \prod_k E_{p_k}(z/\rho_k) \prod_j \Big(E_{p_j}(z/\rho_j)\, E_{p_j}(z/\bar{\rho}_j)\Big)\, h_0(z),$$

with h_0 holomorphic and zero-free, the two factors of each conjugate pair multiplied as displayed. Each primary factor is built from real coefficients, so conjugating it conjugates its argument: $E_p(\bar{w}) = \overline{E_p(w)}$. For a real zero ρ_k the factor $E_{p_k}(z/\rho_k)$ inherits the symmetry in z directly. For a conjugate pair, write $P_j(z) = E_{p_j}(z/\rho_j)\, E_{p_j}(z/\bar{\rho}_j)$ and compute, the two factors exchanging under conjugation:

$$\overline{P_j(\bar{z})} = \overline{E_{p_j}(\bar{z}/\rho_j)}\; \overline{E_{p_j}(\bar{z}/\bar{\rho}_j)} = E_{p_j}(z/\bar{\rho}_j)\, E_{p_j}(z/\rho_j) = P_j(z),$$

real coefficients again, visible at the degree-two core:

$$\big(1 - z/\rho\big)\big(1 - z/\bar{\rho}\big) = 1 - 2\,\mathrm{Re}(1/\rho)\, z + |\rho|^{-2} z^2.$$

Finally, F and every factor satisfy the symmetry, and the symmetry passes to quotients, so the zero-free h_0 satisfies it as well. Every factor of the factorization, h_0 included, is therefore a holomorphic stem with real components.

The lift. The stem formalism of Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2] is stated for real alternative $*$-algebras, hence for $\mathbb{O}$, and for *arbitrary* stems. The real components produced in the previous step give exactly two further properties: the induced section is intrinsic, and on slice-preserving sections the regular product is the pointwise product (Theorem 5.2). So the factorization lifts factor by factor to $A = q^m RSh$ on $\mathbb{O}^* \setminus \{N\}$: R the lift of the real-zero product, S the lift of the paired products, each vanishing exactly on its zero-sphere, and h the lift of h_0, never vanishing. The compactified frame in which this is read, the slice spheres of $\mathbb{O}^*$ with $\varphi_v(\infty) = N$, is GPV's (Remark 3.6; [17, Prop. 4.1, Thm. 4.2]): slice preservation itself carries their facts, before and independently of the transport sections. Convergence transfers because the components are direction-blind: for every I,

$$|A(x+Iy)|^2 = F_1(x+iy)^2 + F_2(x+iy)^2 = |F(x+iy)|^2,$$

so a normal bound for the stem products is the same bound on every slice at once.

Placement in the literature, and the instance used. Over $\mathbb{H}$ the factorization is the theorem of Gentili–Vignozzi [10], with the slice-preserving case in Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2]. Over $\mathbb{O}$ we state and prove it by the reduction above: for an intrinsic section the analysis is the classical theorem at the stem, and the octonionic content is carried entirely by the lift. The instance used in this paper is the A-section's: there the pair R, S is the residue split named in C3, R carrying the residue-$\mathbb{R}$ (classically trivial) zeros and S the residue-$\mathbb{C}$ (classically nontrivial) zero-spheres, the latter infinite in number by C4. □

Lemma 10.2 (Residue-$\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$)**.** *For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of $\mathcal{H}_1$.*

Proof. A is slice-preserving, so on a sphere $\sigma + \gamma S^6$ ($\gamma > 0$) it has the form $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ with α, β real (the real stem, [16]). Suppose A vanishes at one point of the sphere: $\alpha + v\beta = 0$. The real part kills the first component: v is imaginary, so $\mathrm{Re}(\alpha + v\beta) = \alpha = 0$. The modulus of what remains kills the second: $|v\beta| = |v|\,|\beta| = |\beta|$, so $\beta = 0$. Neither component depends on v: the zero occurs at every point of $\sigma + \gamma S^6$ or at none (the spherical-zero structure, [1]); real zeros are isolated real points (residue field $\mathbb{R}$).

The sphere $\sigma + \gamma S^6$ lies in $\mathbb{O}^*$ off the real axis, and it is one G_2-orbit: G_2 acts transitively on S^6 (Theorem 7.5, stabiliser SU(3)), fixing σ and γ and sweeping the direction. The connected components of $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$ are its orbits, the computation $\pi_0(G \ltimes X) = X/G$ (Quillen [22, §1]), so each zero-sphere is one component of $\mathcal{H}_1$. Therefore the residue-$\mathbb{C}$ zeros, the non-real ones, are exactly the 6-sphere components of $A^{-1}(0)$. □

Lemma 10.3 (π_0 of a Grothendieck construction)**.** *For a functor $F : \mathcal{B} \to \mathrm{Grpd}$, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\Big(\int_{\mathcal{B}} F\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}}\,(\pi_0 \circ F).$$

Proof. By Riehl's proof of Lemma 8.3.4, for any $X : \mathcal{C} \to \mathbf{Set}$ each arrow of the category of elements imposes exactly the condition that the corresponding elements be identified in every cone, so that $\pi_0(\mathrm{el}\,X) \cong \mathrm{colim}_{\mathcal{C}}\,X$ [23, p. 102]. Taking $X = \pi_0 \circ F$, the colimit performs exactly the identifications generated by the maps of F. In the Lean development this equivalence is `pi0GrothendieckEquiv`, built from the canonical cocone `pi0Cocone` and `colimit.desc` over Mathlib's `colimit_sound` and `colimit_eq`. □

Lemma 10.4 (Fibrewise π_0 descent and colimit readouts)**.** *Let $F : \mathcal{B} \to \mathrm{Grpd}$ be a functor and let X be a set. Suppose that for every $b \in \mathcal{B}$ a functor*

$$\Lambda_b : F(b) \longrightarrow \mathrm{Disc}(X) \tag{D}$$

has been constructed, and that for every base arrow $h : a \to b$ these functors satisfy

$$\Lambda_a = F(h) \mathbin{⨾} \Lambda_b. \tag{N}$$

Then each Λ_b descends uniquely to a function

$$\lambda_b : \pi_0(F(b)) \longrightarrow X,$$

the functions λ_b assemble into a natural transformation

$$\lambda : \pi_0 \circ F \Longrightarrow \Delta(X), \tag{λ}$$

and the universal property of the colimit produces a unique function

$$\bar{\lambda} : \operatorname*{colim}_{\mathcal{B}}(\pi_0 \circ F) \longrightarrow X \tag{$\bar{\lambda}$}$$

whose composite with the colimit injection from the bth fibre is λ_b. Equivalently, there is a canonical bijection

$$\mathrm{Nat}\big(\pi_0 \circ F, \Delta(X)\big) \;\cong\; \mathrm{Hom}_{\mathbf{Set}}\left(\operatorname*{colim}_{\mathcal{B}}(\pi_0 \circ F), X\right). \tag{U}$$

Here $\mathrm{Disc}(X)$ is the discrete category on the underlying set X, and $\Delta(X)$ is the constant X-valued diagram. Thus X may be any set, including the underlying set of $\mathbb{R}$; no topology on X is used.

Independently, a category J is nonempty and connected exactly when $\pi_0(J)$ is a singleton. Consequently, if $J = \int_{\mathcal{B}} F$ is nonempty and connected, then Lemma 10.3 identifies the component colimit in ($\bar{\lambda}$) with a singleton.

Proof. For each b, the adjunction $\pi_0 \dashv \mathrm{Disc}$ gives

$$\mathrm{Hom}_{\mathbf{Set}}\big(\pi_0(F(b)), X\big) \;\cong\; \mathrm{Hom}_{\mathbf{Cat}}\big(F(b), \mathrm{Disc}(X)\big). \tag{A}$$

This is Riehl's connected-components adjunction ([23, §8.3, Rem. 8.3.5]). In Mathlib the equivalence is

`ConnectedComponents.typeToCatHomEquiv`.

Applying it to Λ_b gives λ_b. Equation (N) says that for every $h : a \to b$ the square

$$\begin{tikzcd}
\pi_0(F(a)) \arrow[r, "\pi_0(F(h))"] \arrow[d, "\lambda_a"'] & \pi_0(F(b)) \arrow[d, "\lambda_b"] \\
X \arrow[r, "1_X"'] & X
\end{tikzcd}$$

commutes. These are precisely the naturality squares for (λ).

A natural transformation $\lambda : \pi_0 \circ F \Rightarrow \Delta(X)$ is a cocone from $\pi_0 \circ F$ to X. A colimit is the universal such cocone: every cocone factors through its colimit cocone by one unique map ([24, §3.1, Defs. 3.1.5–3.1.6]). Applying this universal property to λ gives $\bar{\lambda}$ and the factorization asserted above; varying λ gives the bijection (U). In Mathlib, the induced map is `Limits.colimit.desc`, and its value on each colimit injection is `Limits.colimit.ι_desc` [19].

Finally, Riehl's Remark 8.3.5 gives the remaining criterion. A category is nonempty and connected exactly when its connected-components set is a singleton. Combining that fact with Lemma 10.3 gives the last assertion. ∎

Definition 10.5 (A-sections)**.** An *A-section* is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) with the following four properties.

(C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple and lies at a finite real point p_A, and the pole-cancelled factor $(q - p_A)A(q)$ continues through p_A with a finite nonzero value u_A. This is entirely a finite-point condition.

(C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, A exponentiates one prime-indexed degenerate logarithmic base — a family *made of* the primes, not merely indexed by them: $A = \exp\bigl(\sum_p \ell_p\bigr)$, the sum over the infinite set of primes, where each ℓ_p is slice-preserving, slice-regular, zero-free on Ω_0, and of *prime-power logarithmic form*

$$\ell_p(q) = \sum_{k \geq 1} \frac{a_{p,k}}{k}\, p^{-kq},$$

with real coefficients $a_{p,k}$ bounded uniformly in p and k, the family summable *locally normally* on Ω_0 (the sense in which the cited Euler products converge; Theorem 1.2). In particular A is zero-free on Ω_0; every term decays as $\operatorname{Re} q \to +\infty$, so $A \to 1$ there; and the logarithm of A is *one* exponential base — prime-power frequencies, constant-scale coefficients, no translated anchor, no additive pole term, the pole of C1 arising instead from the divergence of the prime sum at the boundary of the half-space, as for ζ at $s = 1$ — the single base the unique tame continuous lift carries on the degenerate set (Theorems 6.2, 6.5). The prime indexes its term through its own powers, not as a label: each ℓ_p is a power series in p^{-q} for its own prime p, so a family merely enumerated by primes, whose terms are not such power series, is not of this form.

(C3) *Infinite Weierstrass factorization.* Over the *full divisor* of A, including the single pole of (C1), at the real point p_0, multiplicity -1, carried by the explicit left factor, on $\mathbb{O}^* \setminus \{p_0\}$,

$$(q - p_0)\, A \;=\; q^m\, R\, e^g \prod_n \mathcal{E}(\,\cdot\,; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the *nonzero* residue-$\mathbb{R}$ (real) zeros of A, vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue-$\mathbb{C}$ zero-spheres of A, and $\mathcal{E}(\,\cdot\,; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite, the product converging *locally normally* on the finite domain $\mathbb{O} \setminus \{p_0\}$ (the convergence of Proposition 10.1). (The left factor makes "full divisor" literal: the divisor of a meromorphic section includes its single pole; classically, Hadamard factors $(s-1)\zeta(s)$.)

(C4) *Infinitely many residue-$\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$; [2]); residue-$\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Definition 10.6 (The residue-$\mathbb{C}$ subdiagram and its inclusion)**.** By this point conditions C1–C4 have been used to construct the three functors that enter the residue restriction, together with the Grothendieck total of $\mathcal{A}_A$:

$$\begin{gathered} A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld}, \qquad A_{\mathbb{O}} : \mathcal{H}_1 \longrightarrow \mathcal{H}_1, \\ \mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd}, \qquad \mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A. \end{gathered} \tag{A}$$

The first three displayed constructions are functors. The fourth is the Grothendieck total of $\mathcal{A}_A$: $\mathcal{T}_A$ is the category obtained by assembling its fibres over $\mathcal{B}$, and Proposition 9.22 proves that this category is a groupoid. It comes with the canonical projection to $\mathcal{B}$, which sends (b, x) to b and (h, γ) to h: the forgetful functor of the total construction (`CategoryTheory.Grothendieck.forget`; Proposition 9.22).

The functor A^{slice} supplies the positioned slice state at each object of the projective base, and $A_{\mathbb{O}}$ realizes that state by the slice-preserving A-section. Thus an object of the fibre $\mathcal{A}_A(b)$ is the already constructed triple

$$x = \Big((I, u),\ A^{\mathrm{slice}}(b) \cdot (I, u),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b) \cdot (I, u)\big)\Big). \tag{S}$$

The analytic work used to construct these triples has already been done. On the right half-space of C2, where the Euler product has no zeros,

$$\Gamma_A(z) = \sum_p \ell_p(z), \qquad M(\Gamma_A(z)) = \operatorname{diag}\left(\exp\left(\sum_p \ell_p(z)\right), 1\right) = \operatorname{diag}(A(z), 1), \tag{E}$$

each ℓ_p the prime-power series of C2, $\ell_p(z) = \sum_{k\geq 1} \frac{a_{p,k}}{k} p^{-kz}$: the one degenerate base is made of the primes, not merely indexed by them. Condition C1 continues the pole-cancelled factor through its finite real pole p_A, and C3 presents that same factor in Weierstrass form. Evaluation at p_A gives the invertible multiplier from which the A-specific disk action was constructed:

$$\begin{gathered} g_A(z) = (z - p_A) \exp\left(\sum_p \ell_p(z)\right), \\ u_A = g_A(p_A), \qquad D_A = \mathrm{cayleyProjective}\big(\operatorname{diag}(u_A, 1)\big), \\ \exp^{-1}(u_A) = \{\lambda_A + 2\pi i k : k \in \mathbb{Z}\}. \end{gathered} \tag{EW}$$

The last set has the one real level $\operatorname{Re} \lambda_A$, already built into the action total (Remark 9.18). These analytic facts belong to the construction of A^{slice} and $\mathcal{A}_A$. The residue restriction therefore begins with the semantic zero-sphere states that these functors already contain.

The residue subdiagram is defined by two data, both already standing: the semantic C-residue zero locus, which selects the objects, and the natural inclusion into $\mathcal{A}_A$, whose components are the identities of the A-specific fibre groupoids. The hypotheses' one action supplies the arrows and transports in $\mathcal{A}_A$; the residue construction retains them on the selected objects. The naturality square is verified on those object and arrow assignments. This definition states those data. That they define a functor $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ and that the inclusion is full and faithful — fibrewise and on the totals, an inclusion of a subgroupoid — is proved below, at Lemma 10.7.

The selected objects. The semantic C-residue zero locus has already been constructed from the divisor of A:

$$\{z \in \mathbb{C} : A(z) = 0,\ \operatorname{Im} z > 0\}. \tag{Z}$$

By Lemma 10.2, each point of (Z) determines its whole 6-sphere as the G_2-orbit of its slice direction. Condition C3 constructs these zero-sphere states, and condition C4 says that their family is infinite. In the triple (S), the residue condition is the following one statement: the positioned entry

$$A^{\mathrm{slice}}(b) \cdot (I, u)$$

is one of the C-residue zero-sphere states already constructed from (Z). Its determined realization by $A_{\mathbb{O}}$ is the third entry of the same triple.

For a base object a, select a state $x \in \mathcal{A}_A(a)$ precisely when there are an already constructed C-residue zero-sphere state in the fibre at the pole,

$$x_0 \in \mathcal{A}_A(p_A),$$

and a base morphism $h_1 : p_A \to a$ satisfy the concrete action-image equation

$$\mathcal{A}_A(h_1)(x_0) = x. \tag{W}$$

The fibre at p_A is where the hypotheses constructed these states: C1 continues the factor through the pole, and C3 presents it there in Weierstrass form. The morphism h_1 is evaluated by the already constructed functor $\mathcal{A}_A(h_1)$, so its projective/Möbius action and its GPV lift are the A-specific ones already present in $\mathcal{T}_A$. This is the action-category inverse image: the semantic residue condition supplies the zero-sphere states at the pole, and the action of $\mathcal{A}_A$ supplies all of their images over the projective base.

The objects of the residue fibre. Fix a base object a. The action-image equation (W) gives the object collection

$$\begin{aligned} \operatorname{Ob}\mathcal{R}_A(a) := \big\{x \in \operatorname{Ob}\mathcal{A}_A(a) :\ & \text{there are a C-residue zero-sphere state} \\ & x_0 \in \operatorname{Ob}\mathcal{A}_A(p_A) \text{ and an arrow } h_1 : p_A \to a, \\ & \mathcal{A}_A(h_1)(x_0) = x\big\}. \end{aligned} \tag{O}$$

Thus (O) says precisely that $\mathcal{R}_A(a)$ contains the A-specific transports to a of the already-constructed pole-fibre residue states.

One distinction organizes the restriction. The initial zero-sphere states in (W) lie in the one pole fibre; their A-specific transports lie in the other fibres. What distinguishes two initial residue states is their fibre data (I, u): the direction and the normalized input whose positioned image is the state's zero-sphere coordinate, non-real by (Z). The prime-power sum varies along the domain path that built D_A (Definition 9.13), and the zero-sphere coordinates stand as the fixed inputs on which the positioned action acts.

The retained arrows. Let x and y be two objects of the displayed collection. Define

$$\operatorname{Hom}_{\mathcal{R}_A(a)}(x, y) := \big\{A\text{-specific arrows } \gamma : x \to y \text{ in } \mathcal{A}_A(a)\big\}. \tag{H0}$$

This definition places the residue restriction on objects: once the source and target objects are selected, every G_2 arrow between them in the already constructed fibre $\mathcal{A}_A(a)$ is retained.

The identity of a selected object x is the identity 1_x in $\mathcal{A}_A(a)$. Let $\gamma : x \to y$ and $\delta : y \to z$ be retained arrows — $\gamma \cdot x = y$ and $\delta \cdot y = z$ — their composite is the composite in $\mathcal{A}_A(a)$: composition in the action groupoid $\mathcal{A}_A(a)$ is multiplication in G_2 (Definition 9.19), so

$$\gamma \mathbin{⨾} \delta = \delta\gamma :\ x \longrightarrow z, \qquad (\delta\gamma) \cdot x = \delta \cdot (\gamma \cdot x) = \delta \cdot y = z.$$

Its source and target objects x and z are selected, so (H0) places this composite in $\mathcal{R}_A(a)$. Likewise, the inverse of $\gamma : x \to y$ is the group inverse $\gamma^{-1} : y \to x$ in $\mathcal{A}_A(a)$ — inverting the defining equation, $\gamma^{-1} \cdot y = x$ — whose source and target objects are again selected. The category and groupoid equations are the group's own, already proved in $\mathcal{A}_A(a)$:

$$(\gamma \mathbin{⨾} \delta) \mathbin{⨾} \varepsilon = \gamma \mathbin{⨾} (\delta \mathbin{⨾} \varepsilon), \qquad \gamma \mathbin{⨾} \gamma^{-1} = 1_x, \qquad \gamma^{-1} \mathbin{⨾} \gamma = 1_y,$$

associativity of multiplication in G_2 and the two inverse equalities at the relevant objects. Therefore the displayed objects, hom-sets, identities, composites, and inverses construct a groupoid $\mathcal{R}_A(a)$. This constructed groupoid is what is meant by the *full subgroupoid* of $\mathcal{A}_A(a)$ on the selected objects.

The restricted transports. Fix a base arrow $h_2 : a \to b$. On a selected object x of $\mathcal{R}_A(a)$, use the A-specific transport:

$$\mathcal{R}_A(h_2)(x) := \mathcal{A}_A(h_2)(x). \tag{Obj}$$

The right side is again a selected object. Indeed, write the action-image equation for x using $x_0 \in \mathcal{A}_A(p_A)$ and $h_1 : p_A \to a$. Then the composite base morphism $h_1 \mathbin{⨟} h_2 : p_A \to b$ supplies the transported state by functoriality:

$$\begin{aligned} \mathcal{A}_A(h_1 \mathbin{⨟} h_2)(x_0) &= \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x_0)\big) && \text{because } \mathcal{A}_A \text{ preserves composition,} \\ &= \mathcal{A}_A(h_2)(x) && \text{by (W).} \end{aligned}$$

On a retained arrow $\gamma : x \to y$, use the same A-specific transport:

$$\mathcal{R}_A(h_2)(\gamma) := \mathcal{A}_A(h_2)(\gamma), \tag{A}$$

an arrow in $\mathcal{A}_A(b)$ between the two transported objects just shown to be selected, hence retained by (H0). Equations (Obj) and (A) are the A-specific assignments read on the selected states.

The natural inclusion. Define the fibre inclusion $\iota_{A,a} : \mathcal{R}_A(a) \hookrightarrow \mathcal{A}_A(a)$ by

$$\iota_{A,a}(x) := x, \qquad \iota_{A,a}(\gamma) := \gamma. \tag{Inc}$$

Because the operations in $\mathcal{R}_A(a)$ are those of $\mathcal{A}_A(a)$, these two assignments preserve identities and composition and hence define a functor. For a base arrow $h_2 : a \to b$, the naturality square is

$$\begin{tikzcd} \mathcal{R}_A(a) \arrow[r, "\mathcal{R}_A(h_2)"] \arrow[d, hook, "\iota_{A,a}"'] & \mathcal{R}_A(b) \arrow[d, hook, "\iota_{A,b}"] \\ \mathcal{A}_A(a) \arrow[r, "\mathcal{A}_A(h_2)"'] & \mathcal{A}_A(b), \end{tikzcd}$$

Evaluated at an object, this square is equation (Obj); evaluated at an arrow, it is equation (A). On objects both routes apply the same projective/Möbius matrix action $\mathcal{A}_A(h_2)$, and on arrows both routes retain the same G_2 matrix. Thus the two routes agree on every object and arrow. These component squares, one for every h_2, are the natural inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$.

The definition is complete: the locus selects the objects (O), the selected source and target objects retain the arrows (H0), the transports are the restrictions (Obj)–(A), and the inclusion is componentwise the identity (Inc). That these assignments define a functor $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$, and that ι_A is a full and faithful inclusion of a subgroupoid — each fibre inclusion a bijection on hom-sets, and the induced inclusion of totals full and faithful — is Lemma 10.7.

Lemma 10.7 (The residue-$\mathbb{C}$ inclusion)**.** *For an A-section A, the assignments of Definition 10.6 define a functor $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$, the fibre inclusions $\iota_{A,a}$ are full and faithful and assemble into the natural inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$, and the total inclusion*

$$\textstyle\int_{\mathcal{B}} \mathcal{R}_A \xhookrightarrow{\int \iota_A} \int_{\mathcal{B}} \mathcal{A}_A = \mathcal{T}_A$$

is full and faithful: an isomorphism onto its image, which consists exactly of the selected residue states together with the same projective/Möbius transports and G_2 fibre arrows between them.

Proof. Everything is inherited from the already-constructed action functor $\mathcal{A}_A$, after restricting its objects. The closure calculation in Definition 10.6 shows that every $\mathcal{A}_A(h)$ carries selected objects to selected objects, so its restriction $\mathcal{R}_A(h)$ is defined. Since it uses exactly the same object map and the same G_2 matrix on each arrow, the identity calculation is

$$\mathcal{R}_A(1_a)(x) = \mathcal{A}_A(1_a)(x) = x, \qquad \mathcal{R}_A(1_a)(\gamma) = \mathcal{A}_A(1_a)(\gamma) = \gamma,$$

and for $h_2 : a \to b$ and $h_3 : b \to c$ the composite calculation, on an object x, is

$$\mathcal{R}_A(h_2 \mathbin{⨾} h_3)(x) = \mathcal{A}_A(h_2 \mathbin{⨾} h_3)(x) = \mathcal{A}_A(h_3)\big(\mathcal{A}_A(h_2)(x)\big) = \mathcal{R}_A(h_3)\big(\mathcal{R}_A(h_2)(x)\big),$$

with the identical chain on a G_2 matrix γ. Thus identities go to identities, composites go to composites, and $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ is a functor.

For two selected objects x, y of $\mathcal{R}_A(a)$, the map induced by $\iota_{A,a}$ is

$$\begin{aligned} \mathrm{Hom}_{\mathcal{R}_A(a)}(x, y) &\xrightarrow{\ \sim\ } \mathrm{Hom}_{\mathcal{A}_A(a)}\big(\iota_{A,a}x, \iota_{A,a}y\big), \\ \gamma &\longmapsto \iota_{A,a}(\gamma). \end{aligned} \tag{H}$$

The map is injective because it sends each arrow to that same arrow in $\mathcal{A}_A(a)$. It is surjective because (H0) retained every arrow in $\mathcal{A}_A(a)$ between the selected source and target objects. This proves the bijection (H): each $\iota_{A,a}$ is full and faithful.

For a base arrow $h : a \to b$, the square displayed in Definition 10.6 commutes for the same concrete reason. On an object, both routes apply the projective/Möbius matrix action $\mathcal{A}_A(h)$; on a fibre arrow, both retain its G_2 matrix. Therefore the fibre inclusions are the components of the natural transformation $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$. On a selected object x, this says

$$\mathcal{A}_A(h)(\iota_{A,a}x) = \iota_{A,b}\big(\mathcal{R}_A(h)(x)\big). \tag{Src}$$

Now fix (a, x) and (b, y) in the residue total. A morphism between them is exactly a pair consisting of a projective matrix $h : a \to b$ and a G_2 matrix $\gamma : \mathcal{R}_A(h)(x) \to y$ in the fibre at b. By (Src), the induced total inclusion sends this pair to

$$(h, \gamma) \longmapsto \big(h, \iota_{A,b}(\gamma)\big). \tag{Tot}$$

This is faithful: equality after (Tot) first equates the projective matrices h, then the injectivity in (H) equates the G_2 matrices. It is full: given a target pair (h, α), keep its concrete base matrix h; equation (Src) identifies the source of α with the included transported source, and surjectivity in (H) says that this same G_2 matrix is retained in $\mathcal{R}_A(b)$. Hence (h, α) has a preimage. The total inclusion is full and faithful, and its image is exactly the selected residue states with the same A-specific projective/Möbius and G_2 matrix actions between them. ☐

Lemma 10.8 (Transitivity of the residue-$\mathbb{C}$ total)**.** *For an A-section A, the total inverse-image groupoid*

$$\int_{\mathcal{B}} \mathcal{R}_A$$

is transitive: any two of its objects are joined by a morphism.

Proof. Let $P = (a, x)$ and $Q = (b, y)$ be arbitrary objects of $\int_{\mathcal{B}} \mathcal{R}_A$. Their fibre entries are the complete A-section triples

$$
\begin{aligned}
x &= \Big((I_P, u_P),\ A^{\mathrm{slice}}(a)\cdot(I_P,u_P),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(a)\cdot(I_P,u_P)\big)\Big),\\
y &= \Big((I_Q, u_Q),\ A^{\mathrm{slice}}(b)\cdot(I_Q,u_Q),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b)\cdot(I_Q,u_Q)\big)\Big).
\end{aligned}
\tag{T}
$$

We now use the membership data of these two particular objects. By Definition 10.6, there are semantic C-residue zero-sphere states $x_0, y_0 \in \mathcal{A}_A(p_A)$, both in the fibre at the pole — the fibre in which the locus constructs them — and A-specific base morphisms

$$
h_1 : p_A \longrightarrow a, \qquad h_2 : p_A \longrightarrow b, \qquad \mathcal{A}_A(h_1)(x_0) = x, \quad \mathcal{A}_A(h_2)(y_0) = y. \tag{W}
$$

The states x_0 and y_0 are the semantic C-residue zero-sphere states whose transports are the two given objects. Write their complete triples as

$$
\begin{aligned}
x_0 &= \Big((I_1, u_1),\ A^{\mathrm{slice}}(p_A)\cdot(I_1,u_1),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(p_A)\cdot(I_1,u_1)\big)\Big),\\
y_0 &= \Big((I_2, u_2),\ A^{\mathrm{slice}}(p_A)\cdot(I_2,u_2),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(p_A)\cdot(I_2,u_2)\big)\Big).
\end{aligned}
\tag{R}
$$

The middle entries in this display are the positioned states of two members of the already constructed C-residue zero-sphere family. Thus $I_1, I_2 \in S^6$, and their normalized complex coordinates lie in the semantic locus (Z). These displayed entries are exactly their object data in the action groupoid.

The inverse transports in (W) reduce the problem to constructing a morphism inside the one fibre total position at the pole,

$$
(p_A, x_0) \longrightarrow (p_A, y_0). \tag{C}
$$

We construct its base component from the two state-specific evaluations of the one A-specific action. The coordinates u_1 and u_2 lie in the same residue-$\mathbb{C}$ half-plane. Lemma 9.12 therefore gives an invertible matrix morphism $k : p_A \to p_A$ whose action on the normalized coordinate is

$$
A^{\mathrm{slice}}(p_A)^{-1}\cdot\big(A^{\mathrm{slice}}(k)\cdot(A^{\mathrm{slice}}(p_A)\cdot u_1)\big) = u_2. \tag{I}
$$

Every term in this equation is the matrix action of a morphism in the corresponding groupoid category. Equation (I) is exactly the normalized-input entry of Definition 9.20, specialized to the actual arrow $k : p_A \to p_A$. The corresponding square is therefore

$$
\begin{tikzcd}
(I_1, u_1) \arrow[r, "A^{\mathrm{slice}}(p_A)^{-1}A^{\mathrm{slice}}(k)A^{\mathrm{slice}}(p_A)"] \arrow[d, "A^{\mathrm{slice}}(p_A)"'] & (I_1, u_2) \arrow[d, "A^{\mathrm{slice}}(p_A)"] \\
A^{\mathrm{slice}}(p_A)\cdot(I_1,u_1) \arrow[r, "A^{\mathrm{slice}}(k)"'] & A^{\mathrm{slice}}(p_A)\cdot(I_1,u_2).
\end{tikzcd}
\tag{Sq}
$$

Its commutativity is the cancellation already displayed in Definition 9.20, written here on these two particular objects:

$$
\begin{aligned}
A^{\mathrm{slice}}(k)\cdot\big(A^{\mathrm{slice}}(p_A)\cdot u_1\big) &= A^{\mathrm{slice}}(p_A)\cdot\Big(A^{\mathrm{slice}}(p_A)^{-1}\cdot\big(A^{\mathrm{slice}}(k)\cdot(A^{\mathrm{slice}}(p_A)\cdot u_1)\big)\Big)\\
&= A^{\mathrm{slice}}(p_A)\cdot u_2,
\end{aligned}
$$

the first equality being inverse cancellation and the second the concrete input equality (I). Hence the bottom arrow in (Sq) sends the positioned entry of x_0 to the positioned entry of y_0. This is the point at which the proof consumes the Euler–Weierstrass/GPV action already bound into $\mathcal{A}_A$.

Thus $\mathcal{A}_A(k)(x_0)$ and y_0 have the same normalized complex coordinate and the same positioned C-residue coordinate. The remaining comparison in their triples is the slice direction. Since $I_1, I_2 \in S^6$, Theorem 7.5 supplies $\gamma \in G_2$ with $\gamma I_1 = I_2$. It sends the positioned zero-sphere state in direction I_1 to the state with the same complex coordinate in direction I_2. Equivariance of $A_{\mathbb{O}}$ sends the realized third entry by that same element γ. Hence γ is the fibre arrow

$$\gamma : \mathcal{A}_A(k)(x_0) \longrightarrow y_0.$$

The pair (k, γ) is therefore the morphism in (C).

The three stages now compose in the Grothendieck total:

$$P \longrightarrow (p_A, x_0) \xrightarrow{(k,\gamma)} (p_A, y_0) \longrightarrow Q,$$

with base component $(h_1^{-1} \mathbin{⨟} k) \mathbin{⨟} h_2$. Thus the proof of transitivity uses exactly the projective/Möbius arrow k whose action is $\mathcal{A}_A(k)$ and the G_2 arrow γ between its two residue states. Since P and Q were arbitrary, the residue total is transitive. This is where the residue restriction is essential: its realized states are the C-residue zero spheres, one G_2-orbit in each fixed complex coordinate, whereas the ambient total also contains arbitrary realizations in $\mathbb{O}^*$ lying in different G_2-orbits. $\square$

Theorem 10.9 (Concentricity)**.** *Let A be an element of the ring $\mathcal{R}$ of slice-preserving functions on the compactified octonions $\mathbb{O}^*$, satisfying conditions* (C1)–(C4)*. The infinitely many residue-$\mathbb{C}$ zero 6-spheres of A form one connected-component class in the residue action total $\int_{\mathcal{B}} \mathcal{R}_A$; equivalently, their component colimit is a singleton. That single class has one real readout*

$$c \in \mathbb{R}.$$

Every residue-$\mathbb{C}$ zero 6-sphere therefore has centre c, so the infinitely many residue-$\mathbb{C}$ zero spheres are concentric.

Proof. By conditions C1–C4, the following are already constructed:

$$\begin{gathered} A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld}, \qquad A_{\mathbb{O}} : \mathcal{H}_1 \longrightarrow \mathcal{H}_1, \\ \mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd}, \qquad \mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A, \\ \mathcal{R}_A : \mathcal{B} \longrightarrow \mathbf{Grpd}, \qquad \int_{\mathcal{B}} \mathcal{R}_A \xrightarrow{\int \iota_A} \mathcal{T}_A. \end{gathered} \tag{C}$$

The analytic data have already entered these constructions. Condition C2 supplies the prime-power lift and its matrix,

$$\Gamma_A(z) = \sum_p \ell_p(z), \qquad M(\Gamma_A(z)) = \operatorname{diag}(A(z), 1).$$

Condition C1 continues the pole-cancelled function g_A through p_A, and C3 presents that same function in Weierstrass form. At the finite point p_A the resulting multiplier and disk action are

$$g_A(z) = (z - p_A) \exp\left(\sum_p \ell_p(z) \right), \qquad u_A = g_A(p_A), \qquad D_A = \text{cayleyProjective}\big(\operatorname{diag}(u_A, 1)\big). \tag{D}$$

The already-proved functor A^{slice} carries this disk action through the projective base: its object assignment gives the action at each base object and its arrow assignment gives the invertible Möbius matrix action of each base morphism. The functor $A_{\mathbb{O}}$ realizes each positioned state equivariantly, and the action diagram $\mathcal{A}_A$ combines these two functors in the triples

$$\Big((I,u),\ A^{\mathrm{slice}}(b)\cdot(I,u),\ A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b)\cdot(I,u)\big)\Big),$$

whose Grothendieck total is $\mathcal{T}_A$. The GPV lift used in these transports has the exponential fibre

$$\exp^{-1}(u_A)=\{\lambda_A+2\pi ik : k\in\mathbb{Z}\},$$

and every member of that fibre has the same real part $\operatorname{Re}\lambda_A$ (Remark 9.18). Thus the Euler–Weierstrass continuation, the multiplier, and its one real fibre have already been consumed in A^{slice}, $\mathcal{A}_A$, and $\mathcal{T}_A$ before the residue restriction is made.

Condition C3 supplies the semantic C-residue zero-sphere states, and condition C4 makes their family infinite. Definition 10.6 forms the full A-specific inverse-image subdiagram on those states. Lemma 10.7 gives the literal full and faithful inclusion

$$\int_{\mathcal{B}}\mathcal{R}_A \xrightarrow[\text{full and faithful}]{\int\iota_A} \mathcal{T}_A. \tag{I}$$

Its objects are precisely the A-specific images of the displayed zero-sphere triples, and its morphisms are the projective/Möbius and G_2 arrows already constructed in $\mathcal{A}_A$.

Lemma 10.8 constructs a morphism between any two objects of this exact total. Its base component is the locus-specific projective matrix $k : p_A \to p_A$ supplied by the concrete real-projective action (PGL), and the commutative square for the already-constructed functor A^{slice} maps the two positioned coordinates. Its fibre component is the matrix in G_2 that maps the remaining slice direction and, by equivariance of $A_{\mathbb{O}}$, the realized third entry. Hence

$$\int_{\mathcal{B}}\mathcal{R}_A$$

is a transitive groupoid and therefore a connected category.

Connectedness now has two consequences, both for this same residue total. First, Lemma 10.4 says that a nonempty connected category has one connected component. Condition C4 supplies objects, and connectedness was just established; hence

$$\pi_0\Big(\int_{\mathcal{B}}\mathcal{R}_A\Big)$$

is a singleton. Now apply Lemma 10.3 to the specific functor $\mathcal{R}_A : \mathcal{B}\to\mathbf{Grpd}$. The resulting equivalence is

$$\pi_0\Big(\int_{\mathcal{B}}\mathcal{R}_A\Big)\ \cong\ \operatorname*{colim}_{\mathcal{B}}(\pi_0\circ\mathcal{R}_A),$$

so the colimit on the right is the same singleton, which we denote by $\{\kappa\}$.

Only after this collapse do we apply connected components to the residue functor itself:

$$\pi_0\circ\mathcal{R}_A : \mathcal{B}\longrightarrow\mathbf{Set}. \tag{π}$$

For each base object b, the intrinsic real face $\operatorname{Re}\Gamma$ already bound into the A-specific states of $\mathcal{R}_A(b)$ is a functor from that fibre to the discrete category on $\mathbb{R}$. Lemma 10.4, in the form of Mathlib's `ConnectedComponents.typeToCatHomEquiv`, descends this fibrewise functor to a function

$$\lambda_b : \pi_0\big(\mathcal{R}_A(b)\big)\longrightarrow\mathbb{R}.$$

On a class represented by the action-state triple with normalized input (I, u), this is the real coordinate of its positioned entry:

$$\lambda_b([x]) = \operatorname{Re}\Gamma(x) = \operatorname{Re}\Big(A^{\text{slice}}(b) \cdot u\Big). \tag{V}$$

In particular, for the canonical object belonging to the nth C-residue zero sphere, its state equation identifies the positioned entry with $A.\,\text{sphereZero}(n)$, and hence

$$\lambda_b([x_n]) = \operatorname{Re}\big(A.\,\text{sphereZero}(n)\big). \tag{V$_n$}$$

For a base arrow $h : a \to b$, the A-specific action square used by $\mathcal{R}_A(h)$ carries this same bound real face. Consequently the square

$$\begin{tikzcd}
\pi_0\big(\mathcal{R}_A(a)\big) \arrow[r, "\pi_0(\mathcal{R}_A(h))"] \arrow[d, "\lambda_a"'] & \pi_0\big(\mathcal{R}_A(b)\big) \arrow[d, "\lambda_b"] \\
\mathbb{R} \arrow[r, "1_{\mathbb{R}}"'] & \mathbb{R}
\end{tikzcd} \tag{N}$$

commutes. These squares assemble the natural transformation

$$\lambda : \pi_0 \circ \mathcal{R}_A \Longrightarrow \Delta(\mathbb{R}), \tag{λ}$$

where $\Delta(\mathbb{R})$ is the constant diagram on the set $\mathbb{R}$.

The universal property of the colimit now applies to the natural transformation (λ). It produces the one function

$$\bar{\lambda} : \operatorname*{colim}_{\mathcal{B}}(\pi_0 \circ \mathcal{R}_A) \longrightarrow \mathbb{R}. \tag{$\bar{\lambda}$}$$

This is the readout of the component diagram (π). Its relation to the transitivity argument is therefore the explicit chain

$$\pi_0\big(\textstyle\int_{\mathcal{B}} \mathcal{R}_A\big) \xrightarrow{\;\cong\;} \operatorname{colim}_{\mathcal{B}}(\pi_0 \circ \mathcal{R}_A) = \{\kappa\} \xrightarrow{\;\bar{\lambda}\;} \mathbb{R}. \tag{Read}$$

The colimit injection sends every certified class $[x_n]$ to the sole element κ. Apply $\bar{\lambda}$ in (Read) and use the defining calculation (V$_n$). The nth and zeroth representatives therefore give directly

$$\operatorname{Re}\big(A.\,\text{sphereZero}(n)\big) = \bar{\lambda}(\kappa) = \operatorname{Re}\big(A.\,\text{sphereZero}(0)\big). \tag{C$'$}$$

Set $c = \operatorname{Re}\big(A.\,\text{sphereZero}(0)\big)$. Then (C$'$) gives

$$\operatorname{Re}\big(A.\,\text{sphereZero}(n)\big) = c \qquad \text{for every } n. \tag{R}$$

The real coordinate of a residue-$\mathbb{C}$ zero sphere is its centre. Therefore every residue-$\mathbb{C}$ zero 6-sphere has centre c, and the infinitely many such spheres are concentric. □

Corollary 10.10 (Translation to the classical framework)**.** *Under the identification of residue-$\mathbb{C}$ zeros with classically nontrivial zeros, the already-proved residue-$\mathbb{C}$ zero spheres are the classically nontrivial zero spheres and retain the common centre c supplied by Theorem 10.9.*

Proof. This is the translation of the residue-$\mathbb{C}$ family through Theorems 8.1 and 8.2. The centre comes directly from Theorem 10.9, while C4 and Lemma 10.2 supply infinitude and disjointness. □

11 Corollaries

Two corollaries close the argument. The first checks that the octonionic zeta is an A-section, verifying its four conditions C1–C4 one by one against the classical facts of Part 1; the Concentricity Theorem then applies to it directly, and its residue-$\mathbb{C}$ zero-spheres share one real centre c. The second corollary is the Riemann Hypothesis. The functional equation of ξ now pins that centre: it sends a zero of real part c to one of real part $1-c$, and applied to the common centre it forces $c = 1-c$, so $c = \frac{1}{2}$. Every nontrivial zero lies on the critical line.

Corollary 11.1 ($\zeta_{\mathbb{O}^*}$ is an A-section)**.** *$\zeta_{\mathbb{O}^*}$ is an A-section* (Definition 10.5)*: a section of $\mathcal{R}$ satisfying* (C1)–(C4)*.*

Proof. $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1)** *Meromorphic continuation; simple pole.* Classical fact: ζ is meromorphically continued with a single simple pole at $s = 1$ (Theorem 1.1); its pole-cancelled factor $(s-1)\zeta(s)$ continues through $s = 1$ with the finite nonzero value 1. The slice-preserving continuation gives the same statement for $(q-1)\zeta_{\mathbb{O}^*}(q)$.

- **(C2)** *Infinite Euler product.* Classical fact: $\zeta = \prod_p (1-p^{-s})^{-1} = \exp(\sum_p \ell_p)$ on $\Re s > 1$ with $\ell_p(s) = \sum_{k\geq 1} p^{-ks}/k$ — prime-power logarithmic form with every coefficient $a_{p,k} = 1$, one base, decaying to $\zeta \to 1$ as $\Re s \to +\infty$ (Theorem 1.2).

- **(C3)** *Infinite Weierstrass factorization.* Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]): $(q-1)\,\zeta_{\mathbb{O}^*} = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n)$ (pole factor at $p_0 = 1$; classically, Hadamard factors $(s-1)\zeta(s)$), where $m = 0$ ($\zeta(0) = -\frac{1}{2} \neq 0$), R is the product over the nonzero residue-$\mathbb{R}$ zeros, the classically trivial zeros $-2, -4, \ldots$, honest real zeros of $\zeta_{\mathbb{O}^*}$, and $\{q_n\}$ enumerates its residue-$\mathbb{C}$ zero-spheres.

- **(C4)** *Infinitely many residue-$\mathbb{C}$ zeros.* Classical fact: ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue-$\mathbb{C}$ zeros, so $\{q_n\}$ is infinite.

After the (C1)–(C4) verification, the downstream centre-pinning uses the *functional equation* $\xi(s) = \xi(1-s)$, $\xi(\bar{s}) = \overline{\xi(s)}$ (Theorem 1.1), used in Corollary 11.2. The *residue dictionary*, a closed zero of $\zeta_{\mathbb{O}^*}$ is a real point (residue field $\mathbb{R}$, the classically *trivial* zeros) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$, the classically *nontrivial* zeros), is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes "residue-$\mathbb{C}$" and "nontrivial" the same notion. Thus (C1)–(C4) hold. □

Corollary 11.2 (Riemann Hypothesis)**.** *Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.*

Proof. By Corollary 11.1, $\zeta_{\mathbb{O}^*}$ satisfies (C1)–(C4). By Theorem 10.9, its residue-$\mathbb{C}$ value-transport colimit is the real singleton $\{c\}$ and its residue-$\mathbb{C}$ zero spheres are already concentric about c. By Corollary 10.10, these are the classically nontrivial zeros, so every nontrivial zero has real part c. The functional equation $\xi(s) = \xi(1-s)$ (Theorem 1.1) sends a zero of real part c to one of real part $1-c$; applied to the *common* centre it forces $c = 1-c$, whence $c = \frac{1}{2}$ (this is Theorem 8.3 read on $\zeta_{\mathbb{O}^*}$). Therefore every nontrivial zero has real part $\frac{1}{2}$, and by (C4) infinitely many lie on $\Re s = \frac{1}{2}$. □

A word on why the argument takes the shape it does. No one can hold an 8-dimensional graph in the mind's eye, or draw the continuum of Riemann spheres the section lives on. The epigraph names the way through: no single viewpoint sees the whole, but when the viewpoints are multiplied and conjugated, a vision appears that none of them held alone. Each slice direction is one such viewpoint; the value-bearing A-section functor gathers all of them over the base, the total object holds them together, and the colimit conjugates them into a single answer. The concentricity we could not picture becomes one component we can compute.

Remark 11.3 (The image span of a fully faithful inclusion). For any fully faithful functor $F : \mathcal{C} \to \mathcal{D}$ there is a canonical roof through the image,

$$\begin{tikzcd} & \mathrm{Im}(F) \arrow[dl, "F^{-1}"'] \arrow[dr, hook, "j"] & \\ \mathcal{C} & & \mathcal{D} \end{tikzcd}$$

the backward leg being an isomorphism of categories onto the image and the forward leg the canonical inclusion. It is a general fact, available here for ι_A and for $\int \iota_A$, and it is the shape a length-two zigzag argument would use. Theorem 10.9 obtains connectedness directly from transitivity at the total, by a single element. The image span records the corresponding length-two alternative.

Remark 11.4 (Two alternative routes). Two other proofs of Theorem 10.9 are available. The first runs over the base. The restriction sits over $\mathcal{B}$ through its two projections,

$$\begin{tikzcd} \int_{\mathcal{B}} \mathcal{R}_A \arrow[rr, hook, "\int \iota_A"] \arrow[dr, "\pi_{\mathcal{R}}"'] & & \int_{\mathcal{B}} \mathcal{A}_A \arrow[dl, "\pi_{\mathcal{A}}"] \\ & \mathcal{B} & \end{tikzcd}$$

and one may connect residue states footpoint by footpoint: zigzags in the connected base, lifted through the fibres. That route travels through $\mathcal{B}$ and reassembles the value data leg by leg. The second is Thomason's homotopy colimit theorem [25]: the classifying space of a Grothendieck construction is homotopy equivalent to the homotopy colimit of the classifying spaces of its fibres, and reading connected components through that equivalence yields the same collapse of $\pi_0\big(\int_{\mathcal{B}} \mathcal{R}_A\big)$. The proof given here stays at the level of categories, where Lemma 10.3 suffices: the base route assembles what the action already supplies whole, the Thomason route is the topological shadow of the same computation, and both remain expository routes outside the Lean development.

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